

Contents

1 Basic	1
1.1 vimrc	1
1.2 readchar	1
1.3 BigIntIO	1
1.4 Black Magic	1
1.5 Pragma Optimization	2
1.6 Bitset	2
2 Graph	2
2.1 BCC Vertex*	2
2.2 Bridge*	2
2.3 SCC*	2
2.4 2SAT*	2
2.5 MinimumMeanCycle*	2
2.6 Virtual Tree*	3
2.7 Maximum Clique Dyn*	3
2.8 Minimum Steiner Tree*	3
2.9 Dominator Tree*	3
2.10 Four Cycle	4
2.11 Minimum Clique Cover*	4
2.12 Number of Maximal Clique*	4
3 Data Structure	4
3.1 Discrete Trick	4
3.2 BIT kth*	4
3.3 Interval Container*	5
3.4 Leftist Tree	5
3.5 Heavy light Decomposition*	5
3.6 Centroid Decomposition*	5
3.7 LiChaoST*	6
3.8 Link cut tree*	6
3.9 KDTree	6
3.10 Treap	7
4 Flow/Matching	7
4.1 Dinic	7
4.2 Bipartite Matching*	8
4.3 Kuhn Munkres*	8
4.4 MincostMaxflow*	8
4.5 Maximum Simple Graph Matching*	8
4.6 Maximum Weight Matching*	9
4.7 SW-mincut	10
4.8 BoundedFlow*(Dinic*)	10
4.9 Gomory Hu tree*	10
4.10 Minimum Cost Circulation*	10
4.11 Flow Models	11
4.12 matching	11
5 String	11
5.1 KMP	11
5.2 Z-value*	11
5.3 Manacher*	11
5.4 SAIS*	12
5.5 Aho-Corasick Automatan*	12
5.6 Smallest Rotation	12
5.7 De Bruijn sequence*	12
5.8 Extended SAM*	12
5.9 PalTree*	13
5.10 Main Lorentz	13
6 Math	13
6.1 ax+by=gcd(only exgcd*)	13
6.2 Floor and Ceil	13
6.3 Floor Enumeration	13
6.4 Mod Min	13
6.5 Linear Mod Inverse	13
6.6 Linear Filter Mu	14
6.7 Gaussian integer gcd	14
6.8 Gauss Elimination	14
6.9 floor sum*	14
6.10 Miller Rabin*	14

6.11 Fraction	14
6.12 Simultaneous Equations	14
6.13 Pollard Rho*	14
6.14 Simplex Algorithm	14
6.14.1 Construction	15
6.15 Chinese Remainder	15
6.16 Factorial without prime factor*	15
6.17 Pi Count*	15
6.18 Discrete Log*	15
6.19 Berlekamp Massey	16
6.20 Primes	16
6.21 Theorem	16
6.22 Estimation	16
6.23 General Purpose Numbers	16
6.24 Tips for Generating Functions	16
7 Polynomial	16
7.1 Fast Fourier Transform	16
7.2 Number Theory Transform*	17
7.3 Fast Walsh Transform*	17
7.4 Polynomial Operation	17
7.5 Value Polynomial	18
7.6 Newton's Method	18
8 Geometry	18
8.1 Basic	18
8.2 KD Tree	19
8.3 Sector Area	19
8.4 Half Plane Intersection	19
8.5 Rotating Sweep Line	19
8.6 Triangle Center	20
8.7 Polygon Center	20
8.8 Maximum Triangle	20
8.9 Point in Polygon	20
8.10 Circle	20
8.11 Tangent of Circles and Points to Circle	21
8.12 Area of Union of Circles	21
8.13 Minimum Distance of 2 Polygons	21
8.14 2D Convex Hull	21
8.15 3D Convex Hull	22
8.16 Minimum Enclosing Circle	23
8.17 Closest Pair	23
9 Else	23
9.1 Cyclic Ternary Search*	23
9.2 Mo's Algorithm (With modification)	23
9.3 Mo's Algorithm On Tree	23
9.4 Additional Mo's Algorithm Trick	24
9.5 Hilbert Curve	24
9.6 Dynamic Convex Trick*	24
9.7 All LCS*	24
9.8 Adaptive Simpson*	24
9.9 Simulated Annealing	24
9.10 Tree Hash*	24
9.11 Binary Search On Fraction	25
9.12 Min Plus Convolution*	25
9.13 Bitset LCS	25
9.14 N Queens Problem	25
9.15 Digit DP	25
10 Python	25
10.1 Misc	25

"Useful for verifying that there aren't mistypes."

```
ca Hash w !cpp -dD -P -fpreprocessed
  \ | tr -d '[:space:]' \ | md5sum \ | cut -c-6
ca Hash w !cpp -dD -P \ | sed '/^#.*'
  $/d" \ | tr -d '[:space:]' \ | md5sum \ | cut -c-6
ca Hash w !g++ -E - \ | sed '/^#.*'
  $/d" \ | tr -d '[:space:]' \ | md5sum \ | cut -c-6
```

1.2 readchar [a419b9]

```
inline char readchar() {
    static const size_t bufsize = 65536;
    static char buf[bufsize];
    static char *p = buf, *end = buf;
    if (p == end) end = buf +
        fread_unlocked(buf, 1, bufsize, stdin), p = buf;
    return *p++;
}
```

1.3 BigIntIO [d9afcb]

```
__int128 read() {
    __int128 x = 0, f = 1;
    char ch = getchar();
    while (ch < '0' || ch > '9') {
        if (ch == '-') f = -1;
        ch = getchar();
    }
    while (ch >= '0' && ch <= '9') {
        x = x * 10 + ch - '0';
        ch = getchar();
    }
    return x * f;
}
void print(__int128 x) {
    if (x < 0) {
        putchar('-');
        x = -x;
    }
    if (x > 9) print(x / 10);
    putchar(x % 10 + '0');
}
bool cmp(__int128 x, __int128 y) { return x > y; }
```

1.4 Black Magic [0d8b5f]

```
#include <bits/stdc++.h>
#include <bits/extc++.h>
#include <ext/rope>
using namespace __gnu_pbds;
using namespace __gnu_cxx;
#include <ext/pb_ds/assoc_container.hpp>
typedef
    tree<int, null_type, std::less<int>, rb_tree_tag,
        tree_order_statistics_node_update> tree_set;
typedef cc_hash_table<int, int> umap;
typedef priority_queue<int> heap;

int main() {
    // random
    mt19937 rng(chrono::steady_clock::now().time_since_epoch().count());
    int get_rand(int l, int r) { return
        uniform_int_distribution<int>(l, r)(rng); }
    shuffle(v.begin(), v.end(), rng);
    // rb tree
    tree_set s;
    s.insert(71); s.insert(22);
    assert(*s.find_by_order(0) == 22); assert(*s.find_by_order(1) == 71);
    assert(s.order_of_key(22) == 0); assert(s.order_of_key(71) == 1);
    s.erase(22);
    assert(*s.find_by_order(0) == 71); assert(s.order_of_key(71) == 0);
    // mergable heap
    heap a, b; a.join(b);
    // persistent
    rope<char> r[2];
    r[1] = r[0];
    std::string st = "abc";
    r[1].insert(0, st.c_str());
    r[1].erase(1, 1);
    std::cout << r[1].substr(0, 2) << std::endl;
    return 0;
}
```

1 Basic

1.1 vimrc

```
"This file should be placed at ~/.vimrc"
se nu ai hls et ru ic is sc cul
se re=1 ts=4 sts=4 sw=4 ls=2 mouse=a
syntax on
hi cursorline cterm=none ctermbg=89
set bg=dark
inoremap {<CR> {<CR><Esc>ko<tab>
"Select
    region and then type :Hash to hash your selection."
```

1.5 Pragma Optimization [eac636]

```
#pragma GCC optimize("Ofast,no-stack-protector")
#pragma GCC optimize("no-math-errno,unroll-loops")
#pragma GCC target("sse,sse2,sse3,ssse3,sse4")
#pragma GCC target("popcnt,abm,mmx,avx,arch=skylake")
__builtin_ia32_ldmxcsr(__builtin_ia32_stmxcsr())|0x8040
```

1.6 Bitset [cb5d05]

```
#include<bits/stdc++.h>
using namespace std;

int main () {
    bitset<4> bit;
    bit.all(); // all bit is true, ret tru;
    bit.any(); // any bit is true, ret true
    bit.none(); // all bit is false, ret true
    bit.count();
    bit.to_string('0','1');//with parameter
    bit.reset(); // set all to true
    bit.set(); // set all to false
    std::bitset<8> b3{0}, b4{42};
    std::hash<std::bitset<8>> hash_fn8;
    hash_fn8(b3); hash_fn8(b4);
}
```

2 Graph

2.1 BCC Vertex* [740acb]

```
struct BCC { // 0-base
    int n, dft, nbcc;
    vector<int> low, dfn, bln, stk, is_ap, cir;
    vector<vector<int>> G, bcc, nG;
    void make_bcc(int u) {
        bcc.emplace_back(1, u);
        for (; stk.back() != u; stk.pop_back())
            bln[stk.back()] = nbcc, bcc[nbcc].pb(stk.back());
        stk.pop_back(), bln[u] = nbcc++;
    }
    void dfs(int u, int f) {
        int child = 0;
        low[u] = dfn[u] = ++dft, stk.pb(u);
        for (int v : G[u])
            if (!dfn[v]) {
                dfs(v, u), ++child;
                low[u] = min(low[u], low[v]);
                if (dfn[u] <= low[v]) {
                    is_ap[u] = 1, bln[u] = nbcc;
                    make_bcc(v), bcc.back().pb(u);
                }
            } else if (dfn[v] < dfn[u] && v != f)
                low[u] = min(low[u], dfn[v]);
        if (f == -1 && child < 2) is_ap[u] = 0;
        if (f == -1 && child == 0) make_bcc(u);
    }
    BCC(int _n): n(_n), dft(),
        nbcc(), low(n), dfn(n), bln(n), is_ap(n), G(n) {}
    void add_edge(int u, int v) {
        G[u].pb(v), G[v].pb(u);
    }
    void solve() {
        for (int i = 0; i < n; ++i)
            if (!dfn[i]) dfs(i, -1);
    }
    void block_cut_tree() {
        cir.resize(nbcc);
        for (int i = 0; i < n; ++i)
            if (is_ap[i])
                bln[i] = nbcc++;
        cir.resize(nbcc, 1), nG.resize(nbcc);
        for (int i = 0; i < nbcc && !cir[i]; ++i)
            for (int j : bcc[i])
                if (is_ap[j])
                    nG[i].pb(bln[j]), nG[bln[j]].pb(i);
    } // up to 2 * n - 2 nodes!! bln[i] for id
};
```

2.2 Bridge* [4da29a]

```
struct ECC { // 0-base
    int n, dft, ecnt, necc;
    vector<int> low, dfn, bln, is_bridge, stk;
    vector<vector<pii>> G;
    void dfs(int u, int f) {
        dfn[u] = low[u] = ++dft, stk.pb(u);
        for (auto [v, e] : G[u])
```

```
            if (!dfn[v])
                dfs(v, e), low[u] = min(low[u], low[v]);
            else if (e != f)
                low[u] = min(low[u], dfn[v]);
        if (low[u] == dfn[u]) {
            if (f != -1) is_bridge[f] = 1;
            for (; stk.back() != u; stk.pop_back())
                bln[stk.back()] = necc;
            bln[u] = necc++, stk.pop_back();
        }
    }
    ECC(int _n): n(_n), dft(),
        ecnt(), necc(), low(n), dfn(n), bln(n), G(n) {}
    void add_edge(int u, int v) {
        G[u].pb(pii(v, ecnt)), G[v].pb(pii(u, ecnt++));
    }
    void solve() {
        is_bridge.resize(ecnt);
        for (int i = 0; i < n; ++i)
            if (!dfn[i]) dfs(i, -1);
    }
}; // ecc_id(i): bln[i]
// 4da29a
```

2.3 SCC* [4057dc]

```
struct SCC { // 0-base
    int n, dft, nscc;
    vector<int> low, dfn, bln, instack, stk;
    vector<vector<int>> G;
    void dfs(int u) {
        low[u] = dfn[u] = ++dft;
        instack[u] = 1, stk.pb(u);
        for (int v : G[u])
            if (!dfn[v])
                dfs(v), low[u] = min(low[u], low[v]);
            else if (instack[v] && dfn[v] < dfn[u])
                low[u] = min(low[u], dfn[v]);
        if (low[u] == dfn[u]) {
            for (; stk.back() != u; stk.pop_back())
                bln[stk.back()] = nscc, instack[stk.back()] = 0;
            instack[u] = 0, bln[u] = nscc++, stk.pop_back();
        }
    }
    SCC(int _n): n(_n), dft(), nscc(),
        low(n), dfn(n), bln(n), instack(n), G(n) {}
    void add_edge(int u, int v) {
        G[u].pb(v);
    }
    void solve() {
        for (int i = 0; i < n; ++i)
            if (!dfn[i]) dfs(i);
    }
}; // scc_id(i): bln[i]
```

2.4 2SAT* [f5630a]

```
struct SAT { // 0-base
    int n;
    vector<bool> istrue;
    SCC scc;
    SAT(int _n): n(_n), istrue(n + n), scc(n + n) {}
    int rv(int a) {
        return a >= n ? a - n : a + n;
    }
    void add_clause(int a, int b) {
        scc.add_edge(rv(a), b), scc.add_edge(rv(b), a);
    }
    bool solve() {
        scc.solve();
        for (int i = 0; i < n; ++i) {
            if (scc.bln[i] == scc.bln[i + n]) return false;
            istrue[i] = scc.bln[i] < scc.bln[i + n];
            istrue[i + n] = !istrue[i];
        }
        return true;
    }
};
```

2.5 MinimumMeanCycle* [3e5d2b]

```
ll road[N][N]; // input here
struct MinimumMeanCycle {
    ll dp[N + 5][N], n;
    pll solve() {
        ll a = -1, b = -1, L = n + 1;
```

```

for (int i = 2; i <= L; ++i)
    for (int k = 0; k < n; ++k)
        for (int j = 0; j < n; ++j)
            dp[i][j] =
                min(dp[i - 1][k] + road[k][j], dp[i][j]);
for (int i = 0; i < n; ++i) {
    if (dp[L][i] >= INF) continue;
    ll ta = 0, tb = 1;
    for (int j = 1; j < n; ++j)
        if (dp[j][i] < INF &&
            ta * (L - j) < (dp[L][i] - dp[j][i]) * tb)
            ta = dp[L][i] - dp[j][i], tb = L - j;
    if (ta == 0) continue;
    if (a == -1 || a * tb > ta * b) a = ta, b = tb;
}
if (a != -1) {
    ll g = __gcd(a, b);
    return pll(a / g, b / g);
}
return pll(-1LL, -1LL);
}
void init(int _n) {
    n = _n;
    for (int i = 0; i < n; ++i)
        for (int j = 0; j < n; ++j) dp[i + 2][j] = INF;
}
};

```

2.6 Virtual Tree* [1b641b]

```

vector<int> vG[N];
int top, st[N];

void insert(int u) {
    if (top == -1) return st[++top] = u, void();
    int p = LCA(st[top], u);
    if (p == st[top]) return st[++top] = u, void();
    while (top >= 1 && dep[st[top - 1]] >= dep[p])
        vG[st[top - 1]].pb(st[top]), --top;
    if (st[top] != p)
        vG[p].pb(st[top]), --top, st[++top] = p;
    st[++top] = u;
}

void reset(int u) {
    for (int i : vG[u]) reset(i);
    vG[u].clear();
}

void solve(vector<int> &v) {
    top = -1;
    sort(ALL(v),
        [&](int a, int b) { return dfn[a] < dfn[b]; });
    for (int i : v) insert(i);
    while (top > 0) vG[st[top - 1]].pb(st[top]), --top;
    // do something
    reset(v[0]);
}

```

2.7 Maximum Clique Dyn* [d50aa9]

```

struct MaxClique { // fast when N <= 100
    bitset<N> G[N], cs[N];
    int ans, sol[N], q, cur[N], d[N], n;
    void init(int _n) {
        n = _n;
        for (int i = 0; i < n; ++i) G[i].reset();
    }
    void add_edge(int u, int v) {
        G[u][v] = G[v][u] = 1;
    }
    void pre_dfs(vector<int> &r, int l, bitset<N> mask) {
        if (l < 4) {
            for (int i : r) d[i] = (G[i] & mask).count();
            sort(ALL(r),
                [&](int x, int y) { return d[x] > d[y]; });
        }
        vector<int> c(SZ(r));
        int lft = max(ans - q + 1, 1), rgt = 1, tp = 0;
        cs[1].reset(), cs[2].reset();
        for (int p : r) {
            int k = 1;
            while ((cs[k] & G[p]).any()) ++k;
            if (k > rgt) cs[++rgt + 1].reset();
            cs[k][p] = 1;
            if (k < lft) r[tp++] = p;
        }
    }
};

```

```

for (int k = lft; k <= rgt; ++k)
    for (int p = cs[k]._Find_first
        (); p < N; p = cs[k]._Find_next(p))
        r[tp] = p, c[tp] = k, ++tp;
dfs(r, c, l + 1, mask);
}

void dfs(vector<
    int> &r, vector<int> &c, int l, bitset<N> mask) {
    while (!r.empty()) {
        int p = r.back();
        r.pop_back(), mask[p] = 0;
        if (q + c.back() <= ans) return;
        cur[q++] = p;
        vector<int> nr;
        for (int i : r) if (G[p][i]) nr.pb(i);
        if (!nr.empty()) pre_dfs(nr, l, mask & G[p]);
        else if (q > ans) ans = q, copy_n(cur, q, sol);
        c.pop_back(), --q;
    }
}

int solve() {
    vector<int> r(n);
    ans = q = 0, iota(ALL(r), 0);
    pre_dfs(r, 0, bitset<N>(string(n, '1')));
    return ans;
}
};

```

2.8 Minimum Steiner Tree* [62d6fb]

```

struct SteinerTree { // 0-base
    int n, dst[N][N], dp[1 << T][N], tdst[N];
    int vcst[N]; // the cost of vertices
    void init(int _n) {
        n = _n;
        for (int i = 0; i < n; ++i) {
            fill_n(dst[i], n, INF);
            dst[i][i] = vcst[i] = 0;
        }
    }
    void chmin(int &x, int val) {
        x = min(x, val);
    }
    void add_edge(int ui, int vi, int wi) {
        chmin(dst[ui][vi], wi);
    }
    void shortest_path() {
        for (int k = 0; k < n; ++k)
            for (int i = 0; i < n; ++i)
                for (int j = 0; j < n; ++j)
                    chmin(dst[i][j], dst[i][k] + dst[k][j]);
    }
    int solve(const vector<int> &ter) {
        shortest_path();
        int t = SZ(ter), full = (1 << t) - 1;
        for (int i = 0; i <= full; ++i)
            fill_n(dp[i], n, INF);
        copy_n(vkst, n, dp[0]);
        for (int msk = 1; msk <= full; ++msk) {
            if (!(msk & (msk - 1))) {
                int who = __lg(msk);
                for (int i = 0; i < n; ++i)
                    dp[msk][i] = vcst[ter[who]] + dst[ter[who]][i];
            }
            for (int i = 0; i < n; ++i)
                for (int sub = (
                    msk - 1) & msk; sub; sub = (sub - 1) & msk)
                    chmin(dp[msk][i],
                        dp[sub][i] + dp[msk ^ sub][i] - vcst[i]);
            for (int i = 0; i < n; ++i) {
                tdst[i] = INF;
                for (int j = 0; j < n; ++j)
                    chmin(tdst[i], dp[msk][j] + dst[j][i]);
            }
            copy_n(tdst, n, dp[msk]);
        }
        return *min_element(dp[full], dp[full] + n);
    }
}; // O(V 3^T + V^2 2^T)

```

2.9 Dominator Tree* [2b8b32]

```

struct dominator_tree { // 1-base
    vector<int> G[N], rG[N];
    int n, pa[N], dfn[N], id[N], Time;
    int semi[N], idom[N], best[N];
};

```

```

vector<int> tree[N]; // dominator_tree
void init(int _n) {
    n = _n;
    for (int i = 1; i <= n; ++i)
        G[i].clear(), rG[i].clear();
}
void add_edge(int u, int v) {
    G[u].pb(v), rG[v].pb(u);
}
void dfs(int u) {
    id[dfn[u] = ++Time] = u;
    for (auto v : G[u])
        if (!dfn[v]) dfs(v), pa[dfn[v]] = dfn[u];
}
int find(int y, int x) {
    if (y <= x) return y;
    int tmp = find(pa[y], x);
    if (semi[best[y]] > semi[best[pa[y]]])
        best[y] = best[pa[y]];
    return pa[y] = tmp;
}
void tarjan(int root) {
    Time = 0;
    for (int i = 1; i <= n; ++i) {
        dfn[i] = idom[i] = 0;
        tree[i].clear();
        best[i] = semi[i] = i;
    }
    dfs(root);
    for (int i = Time; i > 1; --i) {
        int u = id[i];
        for (auto v : rG[u])
            if (v = dfn[v]) {
                find(v, i);
                semi[i] = min(semi[i], semi[best[v]]);
            }
        tree[semi[i]].pb(i);
        for (auto v : tree[pa[i]]) {
            find(v, pa[i]);
            idom[v] =
                semi[best[v]] == pa[i] ? pa[i] : best[v];
        }
        tree[pa[i]].clear();
    }
    for (int i = 2; i <= Time; ++i) {
        if (idom[i] != semi[i]) idom[i] = idom[idom[i]];
        tree[id[idom[i]]].pb(id[i]);
    }
}
};

```

2.10 Four Cycle [584a52]

```

int main() {
    cin.tie(nullptr)->sync_with_stdio(false);
    cin >> n >> m;
    for (int i = 1; i <= m; ++i) {
        int u, v;
        cin >> u >> v;
        E[u].push_back(v);
        E[v].push_back(u);
        deg[u]++, deg[v]++;
    }
    for (int u = 1; u <= n; u++)
        for (int v : E[u])
            if (deg[u] > deg[v] || (deg[u] == deg[v] && u > v)) E1[u].push_back(v);
    for (int a = 1; a <= n; a++) {
        for (int b : E1[a])
            for (int c : E1[b]) {
                if (deg[a] < deg[c] || (deg[a] == deg[c] && a <= c)) continue;
                total += cnt[c]++;
            }
        for (int b : E1[a])
            for (int c : E1[b]) cnt[c] = 0;
    }
    cout << total << '\n';
    return 0;
}

```

2.11 Minimum Clique Cover* [879472]

```

struct Clique_Cover { // 0-base, O(n2^n)
    int co[1 << N], n, E[N];
    int dp[1 << N];
    void init(int _n) {

```

```

        n = _n, fill_n(dp, 1 << n, 0);
        fill_n(E, n, 0), fill_n(co, 1 << n, 0);
    }
    void add_edge(int u, int v) {
        E[u] |= 1 << v, E[v] |= 1 << u;
    }
    int solve() {
        for (int i = 0; i < n; ++i)
            co[1 << i] = E[i] | (1 << i);
        co[0] = (1 << n) - 1;
        dp[0] = (n & 1) * 2 - 1;
        for (int i = 1; i < (1 << n); ++i) {
            int t = i & -i;
            dp[i] = -dp[i ^ t];
            co[i] = co[i ^ t] & co[t];
        }
        for (int i = 0; i < (1 << n); ++i)
            co[i] = (co[i] & i) == i;
        fwt(co, 1 << n, 1);
        for (int ans = 1; ans < n; ++ans) {
            int sum = 0; // probabilistic
            for (int i = 0; i < (1 << n); ++i)
                sum += (dp[i] * co[i]);
            if (sum) return ans;
        }
        return n;
    }
};

```

2.12 NumberofMaximalClique* [11fa26]

```

struct BronKerbosch { // 1-base
    int n, a[N], g[N][N];
    int S, all[N][N], some[N][N], none[N][N];
    void init(int _n) {
        n = _n;
        for (int i = 1; i <= n; ++i)
            for (int j = 1; j <= n; ++j) g[i][j] = 0;
    }
    void add_edge(int u, int v) {
        g[u][v] = g[v][u] = 1;
    }
    void dfs(int d, int an, int sn, int nn) {
        if (S > 1000) return; // pruning
        if (sn == 0 && nn == 0) ++S;
        int u = some[d][0];
        for (int i = 0; i < sn; ++i) {
            int v = some[d][i];
            if (g[u][v]) continue;
            int tsn = 0, tnn = 0;
            copy_n(all[d], an, all[d + 1]);
            all[d + 1][an] = v;
            for (int j = 0; j < sn; ++j)
                if (g[v][some[d][j]])
                    some[d + 1][tsn++] = some[d][j];
            for (int j = 0; j < nn; ++j)
                if (g[v][none[d][j]])
                    none[d + 1][tnn++] = none[d][j];
            dfs(d + 1, an + 1, tsn, tnn);
            some[d][i] = 0, none[d][nn++] = v;
        }
    }
    int solve() {
        iota(some[0], some[0] + n, 1);
        S = 0, dfs(0, 0, n, 0);
        return S;
    }
};

```

3 Data Structure

3.1 Discrete Trick

```

vector<int> val;
// build
sort(ALL(val)), val.resize(unique(ALL(val)) - val.begin());
// index of x
upper_bound(ALL(val), x) - val.begin();
// max idx <= x
upper_bound(ALL(val), x) - val.begin();
// max idx < x
lower_bound(ALL(val), x) - val.begin();

```

3.2 BIT kth* [e39485]

```

int bit[N + 1]; // N = 2 ^ k

```

```
int query_kth(int k) {
    int res = 0;
    for (int i = N >> 1; i >= 1; i >>= 1)
        if (bit[res + i] < k)
            k -= bit[res += i];
    return res + 1;
}
```

3.3 Interval Container* [c54d29]

```
/* Add and
   remove intervals from a set of disjoint intervals.
   * Will merge the added interval with
   any overlapping intervals in the set when adding.
   * Intervals are [inclusive, exclusive). */
set<pii>::
    iterator addInterval(set<pii>& is, int L, int R) {
        if (L == R) return is.end();
        auto it = is.lower_bound({L, R}), before = it;
        while (it != is.end() && it->X <= R) {
            R = max(R, it->Y);
            before = it = is.erase(it);
        }
        if (it != is.begin() && (--it)->Y >= L) {
            L = min(L, it->X);
            R = max(R, it->Y);
            is.erase(it);
        }
        return is.insert(before, pii(L, R));
    }
void removeInterval(set<pii>& is, int L, int R) {
    if (L == R) return;
    auto it = addInterval(is, L, R);
    auto r2 = it->Y;
    if (it->X == L) is.erase(it);
    else (int&)it->Y = L;
    if (R != r2) is.emplace(R, r2);
}
```

3.4 Leftist Tree [e91538]

```
struct node {
    ll v, data, sz, sum;
    node *l, *r;
    node(ll k)
        : v(0), data(k), sz(1), l(0), r(0), sum(k) {}
};
ll sz(node *p) { return p ? p->sz : 0; }
ll V(node *p) { return p ? p->v : -1; }
ll sum(node *p) { return p ? p->sum : 0; }
node *merge(node *a, node *b) {
    if (!a || !b) return a ? a : b;
    if (a->data < b->data) swap(a, b);
    a->r = merge(a->r, b);
    if (V(a->r) > V(a->l)) swap(a->r, a->l);
    a->v = V(a->r) + 1, a->sz = sz(a->l) + sz(a->r) + 1;
    a->sum = sum(a->l) + sum(a->r) + a->data;
    return a;
}
void pop(node *&o) {
    node *tmp = o;
    o = merge(o->l, o->r);
    delete tmp;
}
```

3.5 Heavy light Decomposition* [b004ae]

```
struct Heavy_light_Decomposition { // 1-base
    int n, ulink[N], deep[N], mxson[N], w[N], pa[N];
    int t, pl[N], data[N], val[N]; // val: vertex data
    vector<int> G[N];
    void init(int _n) {
        n = _n;
        for (int i = 1; i <= n; ++i)
            G[i].clear(), mxson[i] = 0;
    }
    void add_edge(int a, int b) {
        G[a].pb(b), G[b].pb(a);
    }
    void dfs(int u, int f, int d) {
        w[u] = 1, pa[u] = f, deep[u] = d++;
        for (int &i : G[u])
            if (i != f) {
                dfs(i, u, d), w[u] += w[i];
                if (w[mxson[u]] < w[i]) mxson[u] = i;
            }
    }
}
```

```
void cut(int u, int link) {
    data[pl[u] = ++t] = val[u], ulink[u] = link;
    if (!mxson[u]) return;
    cut(mxson[u], link);
    for (int i : G[u])
        if (i != pa[u] && i != mxson[u])
            cut(i, i);
}
void build() { dfs(1, 1, 1), cut(1, 1), /*build*/; }
int query(int a, int b) {
    int ta = ulink[a], tb = ulink[b], res = 0;
    while (ta != tb) {
        if (deep[ta] > deep[tb]) swap(ta, tb), swap(a, b);
        // query(pl[tb], pl[b])
        tb = ulink[b = pa[tb]];
    }
    if (pl[a] > pl[b]) swap(a, b);
    // query(pl[a], pl[b])
}
};
```

3.6 Centroid Decomposition* [5a24da]

```
struct Cent_Dec { // 1-base
    vector<pll> G[N];
    pll info[N]; // store info. of itself
    pll upinfo[N]; // store info. of climbing up
    int n, pa[N], layer[N], sz[N], done[N];
    ll dis[___lg(N) + 1][N];
    void init(int _n) {
        n = _n, layer[0] = -1;
        fill_n(pa + 1, n, 0), fill_n(done + 1, n, 0);
        for (int i = 1; i <= n; ++i) G[i].clear();
    }
    void add_edge(int a, int b, int w) {
        G[a].pb(pll(b, w)), G[b].pb(pll(a, w));
    }
    void get_cent(
        int u, int f, int &mx, int &c, int num) {
        int mxsz = 0;
        sz[u] = 1;
        for (pll e : G[u])
            if (!done[e.X] && e.X != f) {
                get_cent(e.X, u, mx, c, num);
                sz[u] += sz[e.X], mxsz = max(mxsz, sz[e.X]);
            }
        if (mx > max(mxsz, num - sz[u]))
            mx = max(mxsz, num - sz[u]), c = u;
    }
    void dfs(int u, int f, ll d, int org) {
        // if required, add self info or climbing info
        dis[layer[org]][u] = d;
        for (pll e : G[u])
            if (!done[e.X] && e.X != f)
                dfs(e.X, u, d + e.Y, org);
    }
    int cut(int u, int f, int num) {
        int mx = 1e9, c = 0, lc;
        get_cent(u, f, mx, c, num);
        done[c] = 1, pa[c] = f, layer[c] = layer[f] + 1;
        for (pll e : G[c])
            if (!done[e.X]) {
                if (sz[e.X] > sz[c])
                    lc = cut(e.X, c, num - sz[c]);
                else lc = cut(e.X, c, sz[e.X]);
                upinfo[lc] = pll(), dfs(e.X, c, e.Y, c);
            }
        return done[c] = 0, c;
    }
    void build() { cut(1, 0, n); }
    void modify(int u) {
        for (int a = u, ly = layer[a]; a; a = pa[a], --ly) {
            info[a].X += dis[ly][u], ++info[a].Y;
            if (pa[a])
                upinfo[a].X += dis[ly - 1][u], ++upinfo[a].Y;
        }
    }
    ll query(int u) {
        ll rt = 0;
        for (int a = u, ly = layer[a]; a; a = pa[a], --ly) {
            rt += info[a].X + info[a].Y * dis[ly][u];
            if (pa[a])
                rt += upinfo[a].X + upinfo[a].Y * dis[ly - 1][u];
        }
    }
}
```



```

    }
    return rt;
}
};

```

3.7 LiChaoST* [4a4bee]

```

struct L {
    ll m, k, id;
    L() : id(-1) {}
    L(ll a, ll b, ll c) : m(a), k(b), id(c) {}
    ll at(ll x) { return m * x + k; }
};

class LiChao { // maintain max
private:
    int n; vector<L> nodes;
    void insert(int l, int r, int rt, L ln) {
        int m = (l + r) >> 1;
        if (nodes[rt].id == -1)
            return nodes[rt] = ln, void();
        bool atLeft = nodes[rt].at(l) < ln.at(l);
        if (nodes[rt].at(m) < ln.at(m))
            atLeft ^= 1, swap(nodes[rt], ln);
        if (r - l == 1) return;
        if (atLeft) insert(l, m, rt << 1, ln);
        else insert(m, r, rt << 1 | 1, ln);
    }
    ll query(int l, int r, int rt, ll x) {
        int m = (l + r) >> 1; ll ret = -INF;
        if (nodes[rt].id != -1) ret = nodes[rt].at(x);
        if (r - l == 1) return ret;
        if (x < m) return max(ret, query(l, m, rt << 1, x));
        return max(ret, query(m, r, rt << 1 | 1, x));
    }
public:
    LiChao(int n_) : n(n_), nodes(n * 4) {}
    void insert(L ln) { insert(0, n, 1, ln); }
    ll query(ll x) { return query(0, n, 1, x); }
};

```

3.8 Link cut tree* [a35b5d]

```

struct Splay { // xor-sum
    static Splay nil;
    Splay *ch[2], *f;
    int val, sum, rev, size;
    Splay(int _val = 0) : val(_val), sum(_val), rev(0), size(1) {
        f = ch[0] = ch[1] = &nil; }
    bool isr() { return f->ch[0] != this && f->ch[1] != this; }
    int dir() { return f->ch[0] == this ? 0 : 1; }
    void setCh(Splay *c, int d) {
        ch[d] = c;
        if (c != &nil) c->f = this;
        pull();
    }
    void give_tag(int r) {
        if (r) swap(ch[0], ch[1]), rev ^= 1;
    }
    void push() {
        if (ch[0] != &nil) ch[0]->give_tag(rev);
        if (ch[1] != &nil) ch[1]->give_tag(rev);
        rev = 0;
    }
    void pull() {
        // take care of the nil!
        size = ch[0]->size + ch[1]->size + 1;
        sum = ch[0]->sum ^ ch[1]->sum ^ val;
        if (ch[0] != &nil) ch[0]->f = this;
        if (ch[1] != &nil) ch[1]->f = this;
    }
} Splay::nil;
Splay *nil = &Splay::nil;
void rotate(Splay *x) {
    Splay *p = x->f;
    int d = x->dir();
    if (!p->isr()) p->f->setCh(x, p->dir());
    else x->f = p->f;
    p->setCh(x->ch[!d], d);
    x->setCh(p, !d);
    p->pull(), x->pull();
}
void splay(Splay *x) {
    vector<Splay*> splayVec;

```

```

    for (Splay *q = x;; q = q->f) {
        splayVec.pb(q);
        if (q->isr()) break;
    }
    reverse(ALL(splayVec));
    for (auto it : splayVec) it->push();
    while (!x->isr()) {
        if (x->f->isr()) rotate(x);
        else if (x->dir() == x->f->dir())
            rotate(x->f), rotate(x);
        else rotate(x), rotate(x);
    }
}
Splay* access(Splay *x) {
    Splay *q = nil;
    for (; x != nil; x = x->f)
        splay(x), x->setCh(q, 1), q = x;
    return q;
}
void root_path(Splay *x) { access(x), splay(x); }
void chroot(Splay *x) {
    root_path(x), x->give_tag(1);
    x->push(), x->pull();
}
void split(Splay *x, Splay *y) {
    chroot(x), root_path(y);
}
void link(Splay *x, Splay *y) {
    root_path(x), chroot(y);
    x->setCh(y, 1);
}
void cut(Splay *x, Splay *y) {
    split(x, y);
    if (y->size != 5) return;
    y->push();
    y->ch[0] = y->ch[0]->f = nil;
}
Splay* get_root(Splay *x) {
    for (root_path(x); x->ch[0] != nil; x = x->ch[0])
        x->push();
    splay(x);
    return x;
}
bool conn(Splay *x, Splay *y) {
    return get_root(x) == get_root(y);
}
Splay* lca(Splay *x, Splay *y) {
    access(x), root_path(y);
    if (y->f == nil) return y;
    return y->f;
}
void change(Splay *x, int val) {
    splay(x), x->val = val, x->pull();
}
int query(Splay *x, Splay *y) {
    split(x, y);
    return y->sum;
}

```

3.9 KDTree [375ca2]

```

namespace kdt {
    int root, lc[maxn], rc[maxn], xl[maxn], xr[maxn],
        yl[maxn], yr[maxn];
    point p[maxn];
    int build(int l, int r, int dep = 0) {
        if (l == r) return -1;
        function<bool(const point &, const point &> f =
            [dep](const point &a, const point &b) {
                if (dep & 1) return a.x < b.x;
                else return a.y < b.y;
            });
        int m = (l + r) >> 1;
        nth_element(p + l, p + m, p + r, f);
        xl[m] = xr[m] = p[m].x;
        yl[m] = yr[m] = p[m].y;
        lc[m] = build(l, m, dep + 1);
        if (~lc[m]) {
            xl[m] = min(xl[m], xl[lc[m]]);
            xr[m] = max(xr[m], xr[lc[m]]);
            yl[m] = min(yl[m], yl[lc[m]]);
            yr[m] = max(yr[m], yr[lc[m]]);
        }
        rc[m] = build(m + 1, r, dep + 1);
        if (~rc[m]) {
            xl[m] = min(xl[m], xl[rc[m]]);
            xr[m] = max(xr[m], xr[rc[m]]);

```

```

    yl[m] = min(yl[m], yl[rc[m]]);
    yr[m] = max(yr[m], yr[rc[m]]);
}
return m;
}
bool bound(const point &q, int o, long long d) {
    double ds = sqrt(d + 1.0);
    if (q.x < xl[o] - ds || q.x > xr[o] + ds ||
        q.y < yl[o] - ds || q.y > yr[o] + ds)
        return false;
    return true;
}
long long dist(const point &a, const point &b) {
    return (a.x - b.x) * 1ll * (a.x - b.x) +
        (a.y - b.y) * 1ll * (a.y - b.y);
}
void dfs(
    const point &q, long long &d, int o, int dep = 0) {
    if (!bound(q, o, d)) return;
    long long cd = dist(p[o], q);
    if (cd != 0) d = min(d, cd);
    if ((dep & 1) && q.x < p[o].x ||
        !(dep & 1) && q.y < p[o].y) {
        if (~lc[o]) dfs(q, d, lc[o], dep + 1);
        if (~rc[o]) dfs(q, d, rc[o], dep + 1);
    } else {
        if (~rc[o]) dfs(q, d, rc[o], dep + 1);
        if (~lc[o]) dfs(q, d, lc[o], dep + 1);
    }
}
void init(const vector<point> &v) {
    for (int i = 0; i < v.size(); ++i) p[i] = v[i];
    root = build(0, v.size());
}
long long nearest(const point &q) {
    long long res = 1e18;
    dfs(q, res, root);
    return res;
}
} // namespace kdt

```

3.10 Treap [5ab1a1]

```

struct node {
    int data, sz;
    node *l, *r;
    node(int k) : data(k), sz(1), l(0), r(0) {}
    void up() {
        sz = 1;
        if (l) sz += l->sz;
        if (r) sz += r->sz;
    }
    void down() {}
};
int sz(node *a) { return a ? a->sz : 0; }
node *merge(node *a, node *b) {
    if (!a || !b) return a ? a : b;
    if (rand() % (sz(a) + sz(b)) < sz(a))
        return a->down(), a->r = merge(a->r, b), a->up(),
            a;
    return b->down(), b->l = merge(a, b->l), b->up(), b;
}
void split(node *o, node *&a, node *&b, int k) {
    if (!o) return a = b = 0, void();
    o->down();
    if (o->data <= k)
        a = o, split(o->r, a->r, b, k), a->up();
    else b = o, split(o->l, a, b->l, k), b->up();
}
void split2(node *o, node *&a, node *&b, int k) {
    if (sz(o) <= k) return a = o, b = 0, void();
    o->down();
    if (sz(o->l) + 1 <= k)
        a = o, split2(o->r, a->r, b, k - sz(o->l) - 1);
    else b = o, split2(o->l, a, b->l, k);
    o->up();
}
node *kth(node *o, int k) {
    if (k <= sz(o->l)) return kth(o->l, k);
    if (k == sz(o->l) + 1) return o;
    return kth(o->r, k - sz(o->l) - 1);
}
int Rank(node *o, int key) {
    if (!o) return 0;
    if (o->data < key)
        return sz(o->l) + 1 + Rank(o->r, key);
    else return Rank(o->l, key);
}

```

```

}
bool erase(node *&o, int k) {
    if (!o) return 0;
    if (o->data == k) {
        node *t = o;
        o->down(), o = merge(o->l, o->r);
        delete t;
        return 1;
    }
    node *&t = k < o->data ? o->l : o->r;
    return erase(t, k) ? o->up(), 1 : 0;
}
void insert(node *&o, int k) {
    node *a, *b;
    split(o, a, b, k);
    o = merge(a, merge(new node(k), b));
}
void interval(node *&o, int l, int r) {
    node *a, *b, *c;
    split2(o, a, b, l - 1), split2(b, b, c, r);
    // operate
    o = merge(a, merge(b, c));
}

```

4 Flow/Matching

4.1 Dinic [98fb3a]

```

struct MaxFlow { // 0-base
    struct edge {
        int to, cap, flow, rev;
    };
    vector<edge> G[MAXN];
    int s, t, dis[MAXN], cur[MAXN], n;
    int dfs(int u, int cap) {
        if (u == t || !cap) return cap;
        for (int &i = cur[u]; i < (int)G[u].size(); ++i) {
            edge &e = G[u][i];
            if (dis[e.to] == dis[u] + 1 && e.flow != e.cap) {
                int df = dfs(e.to, min(e.cap - e.flow, cap));
                if (df) {
                    e.flow += df;
                    G[e.to][e.rev].flow -= df;
                    return df;
                }
            }
        }
        dis[u] = -1;
        return 0;
    }
    bool bfs() {
        fill_n(dis, n, -1);
        queue<int> q;
        q.push(s), dis[s] = 0;
        while (!q.empty()) {
            int tmp = q.front();
            q.pop();
            for (auto &u : G[tmp])
                if (!dis[u.to] && u.flow != u.cap) {
                    q.push(u.to);
                    dis[u.to] = dis[tmp] + 1;
                }
        }
        return dis[t] != -1;
    }
    int maxflow(int _s, int _t) {
        s = _s, t = _t;
        int flow = 0, df;
        while (bfs()) {
            fill_n(cur, n, 0);
            while ((df = dfs(s, INF))) flow += df;
        }
        return flow;
    }
    void init(int _n) {
        n = _n;
        for (int i = 0; i < n; ++i) G[i].clear();
    }
    void reset() {
        for (int i = 0; i < n; ++i)
            for (auto &j : G[i]) j.flow = 0;
    }
    void add_edge(int u, int v, int cap) {
        G[u].pb(edge{v, cap, 0, (int)G[v].size()});
        G[v].pb(edge{u, 0, 0, (int)G[u].size() - 1});
    }
};

```

4.2 Bipartite Matching* [784535]

```
struct Bipartite_Matching { // 0-base
    int mp[N], mq[N], dis[N+1], cur[N], l, r;
    vector<int> G[N+1];
    bool dfs(int u) {
        for (int &i = cur[u]; i < SZ(G[u]); ++i) {
            int e = G[u][i];
            if (mq[e] == l
                || (dis[mq[e]] == dis[u] + 1 && dfs(mq[e])))
                return mp[mq[e] = u] = e, 1;
        }
        return dis[u] = -1, 0;
    }
    bool bfs() {
        queue<int> q;
        fill_n(dis, l+1, -1);
        for (int i = 0; i < l; ++i)
            if (!mp[i])
                q.push(i), dis[i] = 0;
        while (!q.empty()) {
            int u = q.front();
            q.pop();
            for (int e : G[u])
                if (!dis[mq[e]])
                    q.push(mq[e]), dis[mq[e]] = dis[u] + 1;
        }
        return dis[l] != -1;
    }
    int matching() {
        int res = 0;
        fill_n(mp, l, -1), fill_n(mq, r, l);
        while (bfs()) {
            fill_n(cur, l, 0);
            for (int i = 0; i < l; ++i)
                res += (!mp[i] && dfs(i));
        }
        return res; // (i, mp[i] != -1)
    }
    void add_edge(int s, int t) { G[s].pb(t); }
    void init(int _l, int _r) {
        l = _l, r = _r;
        for (int i = 0; i <= l; ++i)
            G[i].clear();
    }
};
```

4.3 Kuhn Munkres* [4b3863]

```
struct KM { // 0-base, maximum matching
    ll w[N][N], hl[N], hr[N], slk[N];
    int fl[N], fr[N], pre[N], qu[N], ql, qr, n;
    bool vl[N], vr[N];
    void init(int _n) {
        n = _n;
        for (int i = 0; i < n; ++i)
            fill_n(w[i], n, -INF);
    }
    void add_edge(int a, int b, ll wei) {
        w[a][b] = wei;
    }
    bool Check(int x) {
        if (vl[x] == 1, ~fl[x])
            return vr[qu[qr++] = fl[x]] = 1;
        while (~x) swap(x, fr[fl[x] = pre[x]]);
        return 0;
    }
    void bfs(int s) {
        fill_n(slk, n, INF), fill_n(vl, n, 0), fill_n(vr, n, 0);
        ql = qr = 0, qu[qr++] = s, vr[s] = 1;
        for (ll d; ) {
            while (ql < qr)
                for (int x = 0, y = qu[ql++]; x < n; ++x)
                    if (!vl[x] && slk[x] >= (d = hl[x] + hr[y] - w[x][y])) {
                        if (pre[x] == y, d) slk[x] = d;
                        else if (!Check(x)) return;
                    }
            d = INF;
            for (int x = 0; x < n; ++x)
                if (!vl[x] && d > slk[x]) d = slk[x];
            for (int x = 0; x < n; ++x) {
                if (vl[x]) hl[x] += d;
                else slk[x] -= d;
                if (vr[x]) hr[x] -= d;
            }
        }
    }
};
```

```
        for (int x = 0; x < n; ++x)
            if (!vl[x] && !slk[x] && !Check(x)) return;
    }
    ll solve() {
        fill_n(fl, n, -1), fill_n(fr, n, -1), fill_n(hr, n, 0);
        for (int i = 0; i < n; ++i)
            hl[i] = *max_element(w[i], w[i] + n);
        for (int i = 0; i < n; ++i) bfs(i);
        ll res = 0;
        for (int i = 0; i < n; ++i) res += w[i][fl[i]];
        return res;
    }
};
```

4.4 MincostMaxflow* [1c78db]

```
struct MinCostMaxFlow { // 0-base
    struct Edge {
        ll from, to, cap, flow, cost, rev;
    } *past[N];
    vector<Edge> G[N];
    int inq[N], n, s, t;
    ll dis[N], up[N], pot[N];
    bool BellmanFord() {
        fill_n(dis, n, INF), fill_n(inq, n, 0);
        queue<int> q;
        auto relax = [&](int u, ll d, ll cap, Edge *e) {
            if (cap > 0 && dis[u] > d) {
                dis[u] = d, up[u] = cap, past[u] = e;
                if (!inq[u]) inq[u] = 1, q.push(u);
            }
        };
        relax(s, 0, INF, 0);
        while (!q.empty()) {
            int u = q.front();
            q.pop(), inq[u] = 0;
            for (auto &e : G[u]) {
                ll d2 = dis[u] + e.cost + pot[u] - pot[e.to];
                relax(e.to, d2, min(up[u], e.cap - e.flow), &e);
            }
        }
        return dis[t] != INF;
    }
    void solve(int _s, int _t, ll &flow, ll &cost, bool neg = true) {
        s = _s, t = _t, flow = 0, cost = 0;
        if (neg) BellmanFord(), copy_n(dis, n, pot);
        for (; BellmanFord(); copy_n(dis, n, pot)) {
            for (int i = 0; i < n; ++i) dis[i] += pot[i] - pot[s];
            flow += up[t], cost += up[t] * dis[t];
            for (int i = t; past[i]; i = past[i]->from) {
                auto &e = *past[i];
                e.flow += up[t], G[e.to][e.rev].flow -= up[t];
            }
        }
    }
    void init(int _n) {
        n = _n, fill_n(pot, n, 0);
        for (int i = 0; i < n; ++i) G[i].clear();
    }
    void add_edge(ll a, ll b, ll cap, ll cost) {
        G[a].pb(Edge{a, b, cap, 0, cost, SZ(G[b])});
        G[b].pb(Edge{b, a, 0, 0, -cost, SZ(G[a]) - 1});
    }
};
```

4.5 Maximum Simple Graph Matching* [0fe1c3]

```
struct Matching { // 0-base
    queue<int> q; int n;
    vector<int> fa, s, vis, pre, match;
    vector<vector<int>>> G;
    int Find(int u) {
        return u == fa[u] ? u : fa[u] = Find(fa[u]);
    }
    int LCA(int x, int y) {
        static int tk = 0; tk++; x = Find(x); y = Find(y);
        for (; swap(x, y) if (x != n) {
            if (vis[x] == tk) return x;
            vis[x] = tk;
            x = Find(pre[match[x]]);
        }
    }
    void Blossom(int x, int y, int l) {
```



```

    for (; Find(x) != 1; x = pre[y]) {
        pre[x] = y, y = match[x];
        if (s[y] == 1) q.push(y), s[y] = 0;
        for (int z: {x, y}) if (fa[z] == z) fa[z] = 1;
    }
}
bool Bfs(int r) {
    iota(ALL(fa), 0); fill(ALL(s), -1);
    q = queue<int>(); q.push(r); s[r] = 0;
    for (; !q.empty(); q.pop()) {
        for (int x = q.front(); int u : G[x])
            if (s[u] == -1) {
                if (pre[u] = x, s[u] = 1, match[u] == n) {
                    for (int a = u, b = x, last;
                        b != n; a = last, b = pre[a])
                        last =
                            match[b], match[b] = a, match[a] = b;
                    return true;
                }
                q.push(match[u]); s[match[u]] = 0;
            }
        else if (!s[u] && Find(u) != Find(x)) {
            int l = LCA(u, x);
            Blossom(x, u, l); Blossom(u, x, l);
        }
    }
    return false;
}
Matching(int _n) : n(_n), fa(n + 1), s(n + 1), vis
    (n + 1), pre(n + 1, n), match(n + 1, n), G(n) {}
void add_edge(int u, int v)
{ G[u].pb(v), G[v].pb(u); }
int solve() {
    int ans = 0;
    for (int x = 0; x < n; ++x)
        if (match[x] == n) ans += Bfs(x);
    return ans;
} // match[x] == n means not matched
};

```

4.6 Maximum Weight Matching* [lec446]

```

#define REP(i, l, r) for (int i=(l); i<=(r); ++i)
struct WeightGraph { // 1-based
    struct edge { int u, v, w; }; int n, nx;
    vector<int> lab; vector<vector<edge>> g;
    vector<int> slk, match, st, pa, S, vis;
    vector<vector<int>> flo, flo_from; queue<int> q;
    WeightGraph(int n_) : n(n_), nx(n * 2), lab(nx + 1),
        g(nx + 1, vector<edge>(nx + 1)), slk(nx + 1),
        flo(nx + 1, flo_from(nx + 1, vector(n + 1, 0))) {
        match = st = pa = S = vis = slk;
        REP(u, 1, n) REP(v, 1, n) g[u][v] = {u, v, 0};
    }
    int E(edge e)
    { return lab[e.u] + lab[e.v] - g[e.u][e.v].w * 2; }
    void update_slk(int u, int x, int &s)
    { if (!s || E(g[u][x]) < E(g[s][x])) s = u; }
    void set_slk(int x) {
        slk[x] = 0;
        REP(u, 1, n)
            if (g[u][x].w > 0 && st[u] != x && S[st[u]] == 0)
                update_slk(u, x, slk[x]);
    }
    void q_push(int x) {
        if (x <= n) q.push(x);
        else for (int y : flo[x]) q.push(y);
    }
    void set_st(int x, int b) {
        st[x] = b;
        if (x > n) for (int y : flo[x]) set_st(y, b);
    }
    vector<int> split_flo(auto &f, int xr) {
        auto it = find(ALL(f), xr);
        if (auto pr = it - f.begin(); pr % 2 == 1)
            reverse(1 + ALL(f), it = f.end() - pr);
        auto res = vector(f.begin(), it);
        return f.erase(f.begin(), it), res;
    }
    void set_match(int u, int v) {
        match[u] = g[u][v].v;
        if (u <= n) return;
        int xr = flo_from[u][g[u][v].u];
        auto &f = flo[u], z = split_flo(f, xr);
        REP(i, 0, SZ(z) - 1) set_match(z[i], z[i ^ 1]);
        set_match(xr, v); f.insert(f.end(), ALL(z));
    }
    void augment(int u, int v) {

```

```

        for (;;) {
            int xnv = st[match[u]]; set_match(u, v);
            if (!xnv) return;
            set_match(v = xnv, u = st[pa[xnv]]);
        }
    }
    int lca(int u, int v) {
        static int t = 0; ++t;
        for (++t; u || v; swap(u, v)) if (u) {
            if (vis[u] == t) return u;
            vis[u] = t, u = st[match[u]];
            if (u) u = st[pa[u]];
        }
        return 0;
    }
    void add_blossom(int u, int o, int v) {
        int b = find(n + 1 + ALL(st), 0) - begin(st);
        lab[b] = 0, S[b] = 0, match[b] = match[o];
        vector<int> f = {o};
        for (int t : {u, v}) {
            reverse(1 + ALL(f));
            for (int x = t, y; x != o; x = st[pa[y]])
                f.pb(x), f.pb(y = st[match[x]]), q_push(y);
        }
        flo[b] = f; set_st(b, b);
        REP(x, 1, nx) g[b][x].w = g[x][b].w = 0;
        fill(ALL(flo_from[b]), 0);
        for (int xs : flo[b]) {
            REP(x, 1, nx)
                if (g[b][x].w == 0 || E(g[xs][x]) < E(g[b][x]))
                    g[b][x] = g[xs][x], g[x][b] = g[x][xs];
            REP(x, 1, n)
                if (flo_from[xs][x]) flo_from[b][x] = xs;
        }
        set_slk(b);
    }
    void expand_blossom(int b) {
        for (int x : flo[b]) set_st(x, x);
        int xr = flo_from[b][g[b][pa[b]].u], xs = -1;
        for (int x : split_flo(flo[b], xr)) {
            if (xs == -1) { xs = x; continue; }
            pa[xs] = g[x][xs].u, S[xs] = 1, S[x] = 0;
            slk[xs] = 0, set_slk(x), q_push(x), xs = -1;
        }
        for (int x : flo[b])
            if (x == xr) S[x] = 1, pa[x] = pa[b];
            else S[x] = -1, set_slk(x);
        st[b] = 0;
    }
    bool on_found_edge(const edge &e) {
        if (int u = st[e.u], v = st[e.v]; S[v] == -1) {
            int nu = st[match[v]]; pa[v] = e.u; S[v] = 1;
            slk[v] = slk[nu] = S[nu] = 0; q_push(nu);
        }
        else if (S[v] == 0) {
            if (int o = lca(u, v)) add_blossom(u, o, v);
            else return augment(u, v), augment(v, u), true;
        }
        return false;
    }
    bool matching() {
        fill(ALL(S), -1), fill(ALL(slk), 0);
        q = queue<int>();
        REP(x, 1, nx) if (st[x] == x && !match[x])
            pa[x] = S[x] = 0, q_push(x);
        if (q.empty()) return false;
        for (;;) {
            while (SZ(q)) {
                int u = q.front(); q.pop();
                if (S[st[u]] == 1) continue;
                REP(v, 1, n)
                    if (g[u][v].w > 0 && st[u] != st[v]) {
                        if (E(g[u][v]) != 0)
                            update_slk(u, st[v], slk[st[v]]);
                        else if
                            (on_found_edge(g[u][v])) return true;
                    }
            }
            int d = INF;
            REP(b, n + 1, nx) if (st[b] == b && S[b] == 1)
                d = min(d, lab[b] / 2);
            REP(x, 1, nx)
                if (int
                    s = slk[x]; st[x] == x && s && S[x] <= 0)
                    d = min(d, E(g[s][x]) / (S[x] + 2));
            REP(u, 1, n)
                if (S[st[u]] == 1) lab[u] += d;

```

```

    else if (S[st[u]] == 0) {
        if (lab[u] <= d) return false;
        lab[u] -= d;
    }
    REP(b, n + 1, nx) if (st[b] == b && S[b] >= 0)
        lab[b] += d * (2 - 4 * S[b]);
    REP(x, 1, nx)
        if (int s = slk[x]; st[x] == x &&
            s && st[s] != x && E(g[s][x]) == 0)
            if (on_found_edge(g[s][x])) return true;
    REP(b, n + 1, nx)
        if (st[b] == b && S[b] == 1 && lab[b] == 0)
            expand_blossom(b);
}
return false;
}
pair<ll, int> solve() {
    fill(ALL(match), 0);
    REP(u, 0, n) st[u] = u, flo[u].clear();
    int w_max = 0;
    REP(u, 1, n) REP(v, 1, n) {
        flo_from[u][v] = (u == v ? u : 0);
        w_max = max(w_max, g[u][v].w);
    }
    fill(ALL(lab), w_max);
    int n_matches = 0; ll tot_weight = 0;
    while (matching()) ++n_matches;
    REP(u, 1, n) if (match[u] && match[u] < u)
        tot_weight += g[u][match[u]].w;
    return make_pair(tot_weight, n_matches);
}
void add_edge(int u, int v, int w)
{ g[u][v].w = g[v][u].w = w; }
};

```

4.7 SW-mincut [6621f5]

```

struct SW{ // global min cut, O(V^3)
#define REP for (int i = 0; i < n; ++i)
static const int MXN = 514, INF = 2147483647;
int vst[MXN], edge[MXN][MXN], wei[MXN];
void init(int n) {
    REP fill_n(edge[i], n, 0);
}
void addEdge(int u, int v, int w){
    edge[u][v] += w; edge[v][u] += w;
}
int search(int &s, int &t, int n){
    fill_n(vst, n, 0), fill_n(wei, n, 0);
    s = t = -1;
    int mx, cur;
    for (int j = 0; j < n; ++j) {
        mx = -1, cur = 0;
        REP if (wei[i] > mx) cur = i, mx = wei[i];
        vst[cur] = 1, wei[cur] = -1;
        s = t; t = cur;
        REP if (!vst[i]) wei[i] += edge[cur][i];
    }
    return mx;
}
int solve(int n) {
    int res = INF;
    for (int x, y; n > 1; n--){
        res = min(res, search(x, y, n));
        REP edge[i][x] = (edge[x][i] += edge[y][i]);
        REP {
            edge[y][i] = edge[n - 1][i];
            edge[i][y] = edge[i][n - 1];
        } // edge[y][y] = 0;
    }
    return res;
}
} sw;

```

4.8 BoundedFlow*(Dinic*) [4a793f]

```

struct BoundedFlow { // 0-base
    struct edge {
        int to, cap, flow, rev;
    };
    vector<edge> G[N];
    int n, s, t, dis[N], cur[N], cnt[N];
    void init(int _n) {
        n = _n;
        for (int i = 0; i < n + 2; ++i)
            G[i].clear(), cnt[i] = 0;
    }
};

```

```

void add_edge(int u, int v, int lcap, int rcap) {
    cnt[u] -= lcap, cnt[v] += lcap;
    G[u].pb(edge{v, rcap, lcap, SZ(G[v])});
    G[v].pb(edge{u, 0, 0, SZ(G[u]) - 1});
}
void add_edge(int u, int v, int cap) {
    G[u].pb(edge{v, cap, 0, SZ(G[v])});
    G[v].pb(edge{u, 0, 0, SZ(G[u]) - 1});
}
int dfs(int u, int cap) {
    if (u == t || !cap) return cap;
    for (int &i = cur[u]; i < SZ(G[u]); ++i) {
        edge &e = G[u][i];
        if (dis[e.to] == dis[u] + 1 && e.cap != e.flow) {
            int df = dfs(e.to, min(e.cap - e.flow, cap));
            if (df) {
                e.flow += df, G[e.to][e.rev].flow -= df;
                return df;
            }
        }
    }
    dis[u] = -1;
    return 0;
}
bool bfs() {
    fill_n(dis, n + 3, -1);
    queue<int> q;
    q.push(s), dis[s] = 0;
    while (!q.empty()) {
        int u = q.front();
        q.pop();
        for (edge &e : G[u])
            if (!dis[e.to] && e.flow != e.cap)
                q.push(e.to), dis[e.to] = dis[u] + 1;
    }
    return dis[t] != -1;
}
int maxflow(int _s, int _t) {
    s = _s, t = _t;
    int flow = 0, df;
    while (bfs()) {
        fill_n(cur, n + 3, 0);
        while ((df = dfs(s, INF))) flow += df;
    }
    return flow;
}
bool solve() {
    int sum = 0;
    for (int i = 0; i < n; ++i)
        if (cnt[i] > 0)
            add_edge(n + 1, i, cnt[i]), sum += cnt[i];
        else if (cnt[i] < 0) add_edge(i, n + 2, -cnt[i]);
    if (sum != maxflow(n + 1, n + 2)) sum = -1;
    for (int i = 0; i < n; ++i)
        if (cnt[i] > 0)
            G[n + 1].pop_back(), G[i].pop_back();
        else if (cnt[i] < 0)
            G[i].pop_back(), G[n + 2].pop_back();
    return sum != -1;
}
int solve(int _s, int _t) {
    add_edge(_t, _s, INF);
    if (!solve()) return -1; // invalid flow
    int x = G[_t].back().flow;
    return G[_t].pop_back(), G[_s].pop_back(), x;
}
};

```

4.9 Gomory Hu tree* [11be99]

```

MaxFlow Dinic;
int g[MXN];
void GomoryHu(int n) { // 0-base
    fill_n(g, n, 0);
    for (int i = 1; i < n; ++i) {
        Dinic.reset();
        add_edge(i, g[i], Dinic.maxflow(i, g[i]));
        for (int j = i + 1; j <= n; ++j)
            if (g[j] == g[i] && ~Dinic.dis[j])
                g[j] = i;
    }
}

```

4.10 Minimum Cost Circulation* [ba97cf]

```

struct MinCostCirculation { // 0-base
    struct Edge {

```

```

    ll from, to, cap, fcap, flow, cost, rev;
} *past[N];
vector<Edge> G[N];
ll dis[N], inq[N], n;
void BellmanFord(int s) {
    fill_n(dis, n, INF), fill_n(inq, n, 0);
    queue<int> q;
    auto relax = [&](int u, ll d, Edge *e) {
        if (dis[u] > d) {
            dis[u] = d, past[u] = e;
            if (!inq[u]) inq[u] = 1, q.push(u);
        }
    };
    relax(s, 0, 0);
    while (!q.empty()) {
        int u = q.front();
        q.pop(), inq[u] = 0;
        for (auto &e : G[u])
            if (e.cap > e.flow)
                relax(e.to, dis[u] + e.cost, &e);
    }
    void try_edge(Edge &cur) {
        if (cur.cap > cur.flow) return ++cur.cap, void();
        BellmanFord(cur.to);
        if (dis[cur.from] + cur.cost < 0) {
            ++cur.flow, --G[cur.to][cur.rev].flow;
            for (int i = cur.from; past[i]; i = past[i]->from) {
                auto &e = *past[i];
                ++e.flow, --G[e.to][e.rev].flow;
            }
            ++cur.cap;
        }
    }
    void solve(int mxlg) {
        for (int b = mxlg; b >= 0; --b) {
            for (int i = 0; i < n; ++i)
                for (auto &e : G[i])
                    e.cap *= 2, e.flow *= 2;
            for (int i = 0; i < n; ++i)
                for (auto &e : G[i])
                    if (e.fcap >> b & 1)
                        try_edge(e);
        }
    }
    void init(int _n) { n = _n;
        for (int i = 0; i < n; ++i) G[i].clear();
    }
    void add_edge(ll a, ll b, ll cap, ll cost) {
        G[a].pb(Edge{a, b, 0, cap, 0, cost, SZ(G[b]) + (a == b)});
        G[b].pb(Edge{b, a, 0, 0, 0, -cost, SZ(G[a]) - 1});
    }
} mcmf; // O(VE * ElogC)

```

4.11 Flow Models

- Maximum/Minimum flow with lower bound / Circulation problem
 - Constructs supersource S and sink T .
 - For each edge (x, y, l, u) , connect $x \rightarrow y$ with capacity $u - l$.
 - For each vertex v , denote by $in(v)$ the difference between the sum of incoming lower bounds and the sum of outgoing lower bounds.
 - If $in(v) > 0$, connect $S \rightarrow v$ with capacity $in(v)$, otherwise, connect $v \rightarrow T$ with capacity $-in(v)$.
 - To maximize, connect $t \rightarrow s$ with capacity ∞ (skip this in circulation problem), and let f be the maximum flow from S to T . If $f \neq \sum_{v \in V, in(v) > 0} in(v)$, there's no solution. Otherwise, the maximum flow from s to t is the answer.
 - To minimize, let f be the maximum flow from S to T . Connect $t \rightarrow s$ with capacity ∞ and let the flow from S to T be f' . If $f + f' \neq \sum_{v \in V, in(v) > 0} in(v)$, there's no solution. Otherwise, f' is the answer.
 - The solution of each edge e is $l_e + f_e$, where f_e corresponds to the flow of edge e on the graph.
- Construct minimum vertex cover from maximum matching M on bipartite graph (X, Y)
 - Redirect every edge: $y \rightarrow x$ if $(x, y) \in M, x \rightarrow y$ otherwise.
 - DFS from unmatched vertices in X .
 - $x \in X$ is chosen iff x is unvisited.
 - $y \in Y$ is chosen iff y is visited.
- Minimum cost cyclic flow
 - Construct supersource S and sink T
 - For each edge (x, y, c) , connect $x \rightarrow y$ with $(cost, cap) = (c, 1)$ if $c > 0$, otherwise connect $y \rightarrow x$ with $(cost, cap) = (-c, 1)$
 - For each edge with $c < 0$, sum these cost as K , then increase $d(y)$ by 1, decreased $d(x)$ by 1

- For each vertex v with $d(v) > 0$, connect $S \rightarrow v$ with $(cost, cap) = (0, d(v))$
 - For each vertex v with $d(v) < 0$, connect $v \rightarrow T$ with $(cost, cap) = (0, -d(v))$
 - Flow from S to T , the answer is the cost of the flow $C + K$
- Maximum density induced subgraph
 - Binary search on answer, suppose we're checking answer T
 - Construct a max flow model, let K be the sum of all weights
 - Connect sources $s \rightarrow v, v \in G$ with capacity K
 - For each edge (u, v, w) in G , connect $u \rightarrow v$ and $v \rightarrow u$ with capacity w
 - For $v \in G$, connect it with sink $v \rightarrow t$ with capacity $K + 2T - (\sum_{e \in E(v)} w(e)) - 2w(v)$
 - T is a valid answer if the maximum flow $f < K|V|$
 - Minimum weight edge cover
 - For each $v \in V$ create a copy v' , and connect $u' \rightarrow v'$ with weight $w(u, v)$.
 - Connect $v \rightarrow v'$ with weight $2\mu(v)$, where $\mu(v)$ is the cost of the cheapest edge incident to v .
 - Find the minimum weight perfect matching on G' .
 - Project selection problem
 - If $p_v > 0$, create edge (s, v) with capacity p_v ; otherwise, create edge (v, t) with capacity $-p_v$.
 - Create edge (u, v) with capacity w with w being the cost of choosing u without choosing v .
 - The min cut is equivalent to the maximum profit of a subset of projects.
 - Dual of minimum cost maximum flow
 - Capacity c_{uv} , Flow f_{uv} , Cost w_{uv} , Required Flow difference for vertex b_u .
 - If all w_{uv} are integers, then optimal solution can happen when all p_u are integers.

$$\min_{uv} \sum w_{uv} f_{uv} - f_{uv} \geq -c_{uv} \Leftrightarrow \min_u \sum b_u p_u + \sum_{uv} c_{uv} \max(0, p_v - p_u - w_{uv})$$

$$\sum_v f_{vu} - \sum_v f_{uv} = -b_u \quad p_u \geq 0$$

4.12 matching

- 最大匹配 + 最小邊覆蓋 = V
- 最大獨立集 + 最小點覆蓋 = V
- 最大匹配 = 最小點覆蓋
- 最小路徑覆蓋數 = V - 最大匹配數

5 String

5.1 KMP [9bc2e1]

```

int n = s.size(), m = t.size();
vector<int> nxt(m);
for (int i = 1, j = 0; i < m; i++) {
    while (j && t[i] != t[j]) j = nxt[j - 1];
    if (t[i] == t[j]) j++;
    nxt[i] = j;
}
for (int i = 0, j = 0; i < n; i++) {
    while (j && s[i] != t[j]) j = nxt[j - 1];
    if (s[i] == t[j]) j++;
    if (j == m) {
        cout << i - m + 2 << endl;
        j = nxt[j - 1];
    }
}

```

5.2 Z-value* [b47c17]

```

int z[MAXN];
void make_z(const string &s) {
    int l = 0, r = 0;
    for (int i = 1; i < SZ(s); ++i) {
        for (z[i] = max(0, min(r - i + 1, z[i - l]));
             i + z[i] < SZ(s) && s[i + z[i]] == s[z[i]];
             ++z[i]);
        if (i + z[i] - 1 > r) l = i, r = i + z[i] - 1;
    }
}

```

5.3 Manacher* [1ad8ef]

```

int z[MAXN]; // 0-base
/* center i: radius z[i * 2 + 1] / 2
   center i, i + 1: radius z[i * 2 + 2] / 2
   both aba, abba have radius 2 */
void Manacher(string tmp) {
    string s = "%";
    int l = 0, r = 0;
    for (char c : tmp) s.pb(c), s.pb('%');
    for (int i = 0; i < SZ(s); ++i) {
        z[i] = r > i ? min(z[2 * i - l], r - i) : 1;
        while (i - z[i] >= 0 && i + z[i] < SZ(s)
               && s[i + z[i]] == s[i - z[i]]) ++z[i];
    }
}

```

```

    if (z[i] + i > r) r = z[i] + i, l = i;
}
}

```

5.4 SAIS* [6f26bc]

```

auto sais(const auto &s) {
    const int n = SZ(s), z = ranges::max(s) + 1;
    if (n == 1) return vector{0};
    vector<int> c(z); for (int x : s) ++c[x];
    partial_sum(ALL(c), begin(c));
    vector<int> sa(n); auto I = views::iota(0, n);
    vector<bool> t(n, true);
    for (int i = n - 2; i >= 0; --i)
        t[i] = (s[i] == s[i + 1] ? t[i + 1] : s[i] < s[i + 1]);
    auto is_lms = views::filter([&t](int x) {
        return x && t[x] && !t[x - 1];
    });
    auto induce = [&] {
        for (auto x = c; int y : sa)
            if (y-- && (!t[y] || sa[x[s[y] - 1]++] = y);
        for (auto x = c; int y : sa | views::reverse)
            if (y-- && t[y]) sa[--x[s[y]]] = y;
    };
    vector<int> lms, q(n); lms.reserve(n);
    for (auto x = c; int i : I | is_lms)
        q[i] = SZ(lms), lms.pb(sa[-x[s[i]]] = i);
    induce(); vector<int> ns(SZ(lms));
    for (int j = -1, nz = 0; int i : sa | is_lms) {
        if (j >= 0) {
            int len = min({n - i, n - j, lms[q[i] + 1] - i});
            ns[q[i]] = nz += lexicographical_compare(
                begin(s) + j, begin(s) + j + len,
                begin(s) + i, begin(s) + i + len);
        }
        j = i;
    }
    fill(ALL(sa), 0); auto nsa = sais(ns);
    for (auto x = c; int y : nsa | views::reverse)
        y = lms[y], sa[-x[s[y]]] = y;
    return induce(), sa;
}
// sa[i]: sa[i]-th suffix
// is the i-th lexicographically smallest suffix.
// hi[i]: LCP of suffix sa[i] and suffix sa[i - 1].
struct Suffix {
    int n; vector<int> sa, hi, ra;
    Suffix
        (const auto &s, int _n) : n(_n), hi(n), ra(n) {
        vector<int> s(n + 1); // s[n] = 0;
        copy_n(s, n, begin(s)); // s shouldn't contain 0
        sa = sais(s); sa.erase(sa.begin());
        for (int i = 0; i < n; ++i) ra[sa[i]] = i;
        for (int i = 0, h = 0; i < n; ++i) {
            if (!ra[i]) { h = 0; continue; }
            for (int j = sa[ra[i] - 1]; max
                (i, j) + h < n && s[i + h] == s[j + h];) ++h;
            hi[ra[i]] = h ? h - 1 : 0;
        }
    }
};

```

5.5 Aho-Corasick Automatan* [794a77]

```

struct AC_Automatan {
    int nx[len][sigma], fl[len], cnt[len], ord[len], top;
    int rxn[len][sigma]; // node actually be reached
    int newnode() {
        fill_n(nx[top], sigma, -1);
        return top++;
    }
    void init() { top = 1, newnode(); }
    int input(string &s) {
        int X = 1;
        for (char c : s) {
            if (!nx[X][c - 'A']) nx[X][c - 'A'] = newnode();
            X = nx[X][c - 'A'];
        }
        return X; // return the end node of string
    }
    void make_fl() {
        queue<int> q;
        q.push(1), fl[1] = 0;
        for (int t = 0; !q.empty(); ) {
            int R = q.front();
            q.pop(), ord[t++] = R;

```

```

        for (int i = 0; i < sigma; ++i)
            if (~nx[R][i]) {
                int X = rxn[R][i] = nx[R][i], Z = fl[R];
                for (; Z && !nx[Z][i]; ) Z = fl[Z];
                fl[X] = Z ? nx[Z][i] : 1, q.push(X);
            }
        else rxn[R][i] = R > 1 ? rxn[fl[R]][i] : 1;
    }
}
void solve() {
    for (int i = top - 2; i > 0; --i)
        cnt[fl[ord[i]]] += cnt[ord[i]];
}
} ac;

```

5.6 Smallest Rotation [4f469f]

```

string mcp(string s) {
    int n = SZ(s), i = 0, j = 1;
    s += s;
    while (i < n && j < n) {
        int k = 0;
        while (k < n && s[i + k] == s[j + k]) ++k;
        if (s[i + k] <= s[j + k]) j += k + 1;
        else i += k + 1;
        if (i == j) ++j;
    }
    int ans = i < n ? i : j;
    return s.substr(ans, n);
}

```

5.7 De Bruijn sequence* [a09470]

```

constexpr int MAXC = 10, MAXN = 1e5 + 10;
struct DBSeq {
    int C, N, K, L, buf[MAXC * MAXN]; // K <= C^N
    void dfs(int *out, int t, int p, int &ptr) {
        if (ptr >= L) return;
        if (t > N) {
            if (N % p) return;
            for (int i = 1; i <= p && ptr < L; ++i)
                out[ptr++] = buf[i];
        } else {
            buf[t] = buf[t - p], dfs(out, t + 1, p, ptr);
            for (int j = buf[t - p] + 1; j < C; ++j)
                buf[t] = j, dfs(out, t + 1, t, ptr);
        }
    }
    void solve(int _c, int _n, int _k, int *out) {
        int p = 0;
        C = _c, N = _n, K = _k, L = N + K - 1;
        dfs(out, 1, 1, p);
        if (p < L) fill(out + p, out + L, 0);
    }
} dbs;

```

5.8 Extended SAM* [64c3b7]

```

struct exSAM {
    int len[N * 2], link[N * 2]; // maxlength, suflink
    int next[N * 2][CNUM], tot; // [0, tot), root = 0
    int lenSorted[N * 2]; // topo. order
    int cnt[N * 2]; // occurrence
    int newnode() {
        fill_n(next[tot], CNUM, 0);
        len[tot] = cnt[tot] = link[tot] = 0;
        return tot++;
    }
    void init() { tot = 0, newnode(), link[0] = -1; }
    int insertSAM(int last, int c) {
        int cur = next[last][c];
        len[cur] = len[last] + 1;
        int p = link[last];
        while (p != -1 && !next[p][c])
            next[p][c] = cur, p = link[p];
        if (p == -1) return link[cur] = 0, cur;
        int q = next[p][c];
        if (len
            [p] + 1 == len[q]) return link[cur] = q, cur;
        int clone = newnode();
        for (int i = 0; i < CNUM; ++i)
            next[
                clone][i] = len[next[q][i]] ? next[q][i] : 0;
        len[clone] = len[p] + 1;
        while (p != -1 && next[p][c] == q)
            next[p][c] = clone, p = link[p];
        link[link[cur] = clone] = link[q];
    }

```

```

    link[q] = clone;
    return cur;
}
void insert(const string &s) {
    int cur = 0;
    for (auto ch : s) {
        int &nxt = next[cur][int(ch - 'a')];
        if (!nxt) nxt = newnode();
        cnt[cur = nxt] += 1;
    }
}
void build() {
    queue<int> q;
    q.push(0);
    while (!q.empty()) {
        int cur = q.front();
        q.pop();
        for (int i = 0; i < CNUM; ++i)
            if (next[cur][i])
                q.push(insertSAM(cur, i));
    }
    vector<int> lc(tot);
    for (int i = 1; i < tot; ++i) ++lc[len[i]];
    partial_sum(ALL(lc), lc.begin());
    for (int i = 1; i < tot; ++i) lenSorted[--lc[len[i]]] = i;
}
void solve() {
    for (int i = tot - 2; i >= 0; --i)
        cnt[link[lenSorted[i]]] += cnt[lenSorted[i]];
}
};

```

5.9 PalTree* [d7d2cf]

```

struct palindromic_tree {
    struct node {
        int next[26], fail, len;
        int cnt, num; // cnt: appear times, num: number of
                       // pal. suf.
        node(int l = 0) : fail(0), len(l), cnt(0), num(0) {
            for (int i = 0; i < 26; ++i) next[i] = 0;
        }
    };
    vector<node> St;
    vector<char> s;
    int last, n;
    palindromic_tree() : St(2), last(1), n(0) {
        St[0].fail = 1, St[1].len = -1, s.pb(-1);
    }
    inline void clear() {
        St.clear(), s.clear(), last = 1, n = 0;
        St.pb(0), St.pb(-1);
        St[0].fail = 1, s.pb(-1);
    }
    inline int get_fail(int x) {
        while (s[n - St[x].len - 1] != s[n])
            x = St[x].fail;
        return x;
    }
    inline void add(int c) {
        s.push_back(c - 'a'), ++n;
        int cur = get_fail(last);
        if (!St[cur].next[c]) {
            int now = SZ(St);
            St.pb(St[cur].len + 2);
            St[now].fail =
                St[get_fail(St[cur].fail)].next[c];
            St[cur].next[c] = now;
            St[now].num = St[St[now].fail].num + 1;
        }
        last = St[cur].next[c], ++St[last].cnt;
    }
    inline void count() { // counting cnt
        auto i = St.rbegin();
        for (; i != St.rend(); ++i) {
            St[i->fail].cnt += i->cnt;
        }
    }
    inline int size() { // The number of diff. pal.
        return SZ(St) - 2;
    }
};

```

5.10 Main Lorentz [615b8f]

```
vector<pair<int, int>> rep[kN]; // 0-base [1, r]
```

```

void main_lorentz(const string &s, int sft = 0) {
    const int n = s.size();
    if (n == 1) return;
    const int nu = n / 2, nv = n - nu;
    const string u = s.substr(0, nu), v = s.substr(nu,
        ru(u.rbegin(), u.rend()), rv(v.rbegin(), v.rend()));
    main_lorentz(u, sft), main_lorentz(v, sft + nu);
    const auto z1 = Zalgo(ru), z2 = Zalgo(v + '#' + u),
        z3 = Zalgo(ru + '#' + rv), z4 = Zalgo(v);
    auto get_z = [](const vector<int> &z, int i) {
        return
            (0 <= i and i < (int)z.size()) ? z[i] : 0;
    };
    auto add_rep
        = [&](bool left, int c, int l, int k1, int k2) {
            const
                int L = max(1, l - k2), R = min(l - left, k1);
            if (L > R) return;
            if (left)
                rep[l].emplace_back(sft + c - R, sft + c - L);
            else
                rep[l].emplace_back
                    (sft + c - R - l + 1, sft + c - L - l + 1);
        };
    for (int cntr = 0; cntr < n; cntr++) {
        int l, k1, k2;
        if (cntr < nu) {
            l = nu - cntr;
            k1 = get_z(z1, nu - cntr);
            k2 = get_z(z2, nv + 1 + cntr);
        } else {
            l = cntr - nu + 1;
            k1 = get_z(z3, nu + 1 + nv - 1 - (cntr - nu));
            k2 = get_z(z4, (cntr - nu) + 1);
        }
        if (k1 + k2 >= l)
            add_rep(cntr < nu, cntr, l, k1, k2);
    }
} // p \in [1, r] => s[p, p + i) = s[p + i, p + 2i)

```

6 Math

6.1 ax+by=gcd(only exgcd *) [7b833d]

```

pll exgcd(ll a, ll b) {
    if (b == 0) return pll(1, 0);
    ll p = a / b;
    pll q = exgcd(b, a % b);
    return pll(q.Y, q.X - q.Y * p);
}
/* ax+by=res, let x be minimum non-negative
g, p = gcd(a, b), exgcd(a, b) * res / g
if p.X < 0: t = (abs(p.X) + b / g - 1) / (b / g)
else: t = -(p.X / (b / g))
p += (b / g, -a / g) * t */

```

6.2 Floor and Ceil [692c04]

```

int floor(int a, int b)
{ return a / b - (a % b && (a < 0) ^ (b < 0)); }
int ceil(int a, int b)
{ return a / b + (a % b && (a < 0) ^ (b > 0)); }

```

6.3 Floor Enumeration [7cbcdf]

```

// enumerating x = floor(n / i), [1, r]
for (int l = 1, r; l <= n; l = r + 1) {
    int x = n / l;
    r = n / x;
}

```

6.4 Mod Min [9118e1]

```

// min{k | 1 <= ((ak) mod m) <= r}, no solution -> -1
ll mod_min(ll a, ll m, ll l, ll r) {
    if (a == 0) return l ? -1 : 0;
    if (ll k = (l + a - 1) / a; k * a <= r)
        return k;
    ll b = m / a, c = m % a;
    if (ll y = mod_min(c, a, a - r % a, a - l % a))
        return (l + y * c + a - 1) / a + y * b;
    return -1;
}

```

6.5 Linear Mod Inverse [5a4cbf]

```

inv[1] = 1;
for (int i = 2; i
    <= N; ++i) inv[i] = ((mod - mod / i) * inv[mod % i]) % mod;

```


6.6 Linear Filter Mu [ac2ac3]

```
void getMu() {
    mu[1] = 1;
    for (int i = 2; i <= n; ++i) {
        if (!flg[i]) p[++tot] = i, mu[i] = -1;
        for (int j = 1; j <= tot && i * p[j] <= n; ++j) {
            flg[i * p[j]] = 1;
            if (i % p[j] == 0) {
                mu[i * p[j]] = 0;
                break;
            }
            mu[i * p[j]] = -mu[i];
        }
    }
}
```

6.7 Gaussian integer gcd [763e59]

```
cpx gaussian_gcd(cpx a, cpx b) {
#define rnd
    (a, b) ((a >= 0 ? a * 2 + b : a * 2 - b) / (b * 2))
    ll c = a.real() * b.real() + a.imag() * b.imag();
    ll d = a.imag() * b.real() - a.real() * b.imag();
    ll r = b.real() * b.real() + b.imag() * b.imag();
    if (c % r == 0 && d % r == 0) return b;
    return gaussian_gcd
        (b, a - cpx(rnd(c, r), rnd(d, r)) * b);
}
```

6.8 GaussElimination [6308be]

```
void GAS(V<V<double>>&vc) {
    int len = vc.size();
    for (int i = 0; i < len; ++i) {
        int idx = find_if(vc.begin() + i, vc.end(), [&](
            auto&v) { return v[i] != 0; }) - vc.begin();
        if (idx == len) continue;
        if (i != idx) swap(vc[idx], vc[i]);
        double pivot = vc[i][i];
        for_each(vc[i].begin(), vc[i].end(), [&](auto &a) { a /= pivot; });
        for (int j = 0; j < len; ++j) {
            if (i == j) continue;
            if (vc[j][i] != 0) {
                double mul = vc[j][i] / vc[i][i];
                transform(vc[j].begin(), vc[j].end(),
                    vc[i].begin(), vc[j].begin(),
                    [&](auto &a, auto &b) {
                        return a - b * mul;
                    });
            }
        }
    }
};
```

6.9 floor sum* [49de67]

```
ll floor_sum(ll n, ll m, ll a, ll b) {
    ll ans = 0;
    if (a >= m)
        ans += (n - 1) * n * (a / m) / 2, a %= m;
    if (b >= m)
        ans += n * (b / m), b %= m;
    ll y_max
        = (a * n + b) / m, x_max = (y_max * m - b);
    if (y_max == 0) return ans;
    ans += (n - (x_max + a - 1) / a) * y_max;
    ans += floor_sum(y_max, a, m, (a - x_max % a) % a);
    return ans;
} // sum^{n-1}_0 floor((a * i + b) / m) in log(n + m + a + b)
```

6.10 Miller Rabin* [06308c]

```
// n < 4,759,123,141      3 : 2, 7, 61
// n < 1,122,004,669,633  4 : 2, 13, 23, 1662803
// n < 3,474,749,660,383  6 : primes <= 13
// n < 2^64               7 :
// 2, 325, 9375, 28178, 450775, 9780504, 1795265022
bool Miller_Rabin(ll a, ll n) {
    if ((a = a % n) == 0) return 1;
    if (n % 2 == 0) return n == 2;
    ll tmp = (n - 1) / ((n - 1) & (1 - n));
    ll t = __lg(((n - 1) & (1 - n))), x = 1;
    for (; tmp; tmp >>= 1, a = mul(a, a, n))
        if (tmp & 1) x = mul(x, a, n);
    if (x == 1 || x == n - 1) return 1;
```

```
while (--t)
    if ((x = mul(x, x, n)) == n - 1) return 1;
return 0;
}
```

6.11 Fraction [4ab37a]

```
struct fraction {
    ll n, d;
    fraction
        (const ll &n=0, const ll &d=1): n(_n), d(_d) {
        ll t = gcd(n, d);
        n /= t, d /= t;
        if (d < 0) n = -n, d = -d;
    }
    fraction operator-() const
    { return fraction(-n, d); }
    fraction operator+(const fraction &b) const
    { return fraction(n * b.d + b.n * d, d * b.d); }
    fraction operator-(const fraction &b) const
    { return fraction(n * b.d - b.n * d, d * b.d); }
    fraction operator*(const fraction &b) const
    { return fraction(n * b.n, d * b.d); }
    fraction operator/(const fraction &b) const
    { return fraction(n * b.d, d * b.n); }
    void print() {
        cout << n;
        if (d != 1) cout << "/" << d;
    }
};
```

6.12 Simultaneous Equations [a231be]

```
struct matrix { //m variables, n equations
    int n, m;
    fraction M[MAXN][MAXN + 1], sol[MAXN];
    int solve() { // -1: inconsistent, >= 0: rank
        for (int i = 0; i < n; ++i) {
            int piv = 0;
            while (piv < m && !M[i][piv].n) ++piv;
            if (piv == m) continue;
            for (int j = 0; j < n; ++j) {
                if (i == j) continue;
                fraction tmp = -M[j][piv] / M[i][piv];
                for (int k = 0; k <=
                    m; ++k) M[j][k] = tmp * M[i][k] + M[j][k];
            }
        }
        int rank = 0;
        for (int i = 0; i < n; ++i) {
            int piv = 0;
            while (piv < m && !M[i][piv].n) ++piv;
            if (piv == m && M[i][m].n) return -1;
            else if (piv
                < m) ++rank, sol[piv] = M[i][m] / M[i][piv];
        }
        return rank;
    }
};
```

6.13 Pollard Rho* [fdef9b]

```
map<ll, int> cnt;
void PollardRho(ll n) {
    if (n == 1) return;
    if (prime(n)) return ++cnt[n], void();
    if (n % 2
        == 0) return PollardRho(n / 2), ++cnt[2], void();
    ll x = 2, y = 2, d = 1, p = 1;
#define f(x, n, p) ((mul(x, x, n) + p) % n)
    while (true) {
        if (d != n && d != 1) {
            PollardRho(n / d);
            PollardRho(d);
            return;
        }
        if (d == n) ++p;
        x = f(x, n, p), y = f(f(y, n, p), n, p);
        d = gcd(abs(x - y), n);
    }
}
```

6.14 Simplex Algorithm [6b4566]

```
const int MAXN = 11000, MAXM = 405;
const double eps = 1E-10;
double a[MAXN][MAXM], b[MAXN], c[MAXN];
double d[MAXN][MAXM], x[MAXN];
```

```

int ix[MAXN + MAXM]; // !!! array all indexed from 0
// max{cx} subject to {Ax<=b,x>=0}
// n: constraints, m: vars !!!
// x[] is the optimal solution vector
// usage :
// value = simplex(a, b, c, N, M);
double simplex(int n, int m){
    +hm;
    fill_n(d[n], m + 1, 0);
    fill_n(d[n + 1], m + 1, 0);
    iota(ix, ix + n + m, 0);
    int r = n, s = m - 1;
    for (int i = 0; i < n; ++i) {
        for (int j = 0; j < m - 1; ++j) d[i][j] = -a[i][j];
        d[i][m - 1] = 1;
        d[i][m] = b[i];
        if (d[r][m] > d[i][m]) r = i;
    }
    copy_n(c, m - 1, d[n]);
    d[n + 1][m - 1] = -1;
    for (double dd;; ) {
        if (r < n) {
            swap(ix[s], ix[r + m]);
            d[r][s] = 1.0 / d[r][s];
            for (int j = 0; j <= m; ++j)
                if (j != s) d[r][j] *= -d[r][s];
            for (int i = 0; i <= n + 1; ++i) if (i != r) {
                for (int j = 0; j <= m; ++j) if (j != s)
                    d[i][j] += d[r][j] * d[i][s];
                d[i][s] *= d[r][s];
            }
        }
        r = s = -1;
        for (int j = 0; j < m; ++j)
            if (s < 0 || ix[s] > ix[j]) {
                if (d[n + 1][j] > eps ||
                    (d[n + 1][j] > -eps && d[n][j] > eps))
                    s = j;
            }
        if (s < 0) break;
        for (int i = 0; i < n; ++i) if (d[i][s] < -eps) {
            if (r < 0 ||
                (dd = d[r][m] / d[r][s] - d[i][m] / d[i][s]) < -eps ||
                (dd < eps && ix[r + m] > ix[i + m]))
                r = i;
        }
        if (r < 0) return -1; // not bounded
    }
    if (d[n + 1][m] < -eps) return -1; // not executable
    double ans = 0;
    fill_n(x, m, 0);
    for (int i = m; i < n + m; ++i) { // the missing enumerated x[i] = 0
        if (ix[i] < m - 1) {
            ans += d[i - m][m] * c[ix[i]];
            x[ix[i]] = d[i - m][m];
        }
    }
    return ans;
}

```

6.14.1 Construction

Primal	Dual
Maximize $c^T x$ s.t. $Ax \leq b, x \geq 0$	Minimize $b^T y$ s.t. $A^T y \geq c, y \geq 0$
Maximize $c^T x$ s.t. $Ax \leq b$	Minimize $b^T y$ s.t. $A^T y = c, y \geq 0$
Maximize $c^T x$ s.t. $Ax = b, x \geq 0$	Minimize $b^T y$ s.t. $A^T y \geq c$

$\bar{x}$ and $\bar{y}$ are optimal if and only if for all $i \in [1, n]$, either $\bar{x}_i = 0$ or $\sum_{j=1}^m A_{ji} \bar{y}_j = c_i$

holds and for all $i \in [1, m]$ either $\bar{y}_i = 0$ or $\sum_{j=1}^n A_{ij} \bar{x}_j = b_j$ holds.

1. In case of minimization, let $c'_i = -c_i$
2. $\sum_{1 \leq i \leq n} A_{ji} x_i \geq b_j \rightarrow \sum_{1 \leq i \leq n} -A_{ji} x_i \leq -b_j$
3. $\sum_{1 \leq i \leq n} A_{ji} x_i = b_j$
 - $\sum_{1 \leq i \leq n} A_{ji} x_i \leq b_j$
 - $\sum_{1 \leq i \leq n} A_{ji} x_i \geq b_j$

4. If x_i has no lower bound, replace x_i with $x_i - x'_i$

6.15 chineseRemainder [a53b6d]

```

ll solve(ll x1, ll m1, ll x2, ll m2) {
    ll g = gcd(m1, m2);
    if ((x2 - x1) % g) return -1; // no sol
    m1 /= g; m2 /= g;
    pll p = exgcd(m1, m2);
    ll lcm = m1 * m2 * g;
    ll res = p.first * (x2 - x1) * m1 + x1;
    // be careful with overflow
}

```

```

return (res % lcm + lcm) % lcm;
}

```

6.16 Factorial without prime factor* [c324f3]

```

// O(p^k + log^2 n), pk = p^k
ll prod[MAXP];
ll fac_no_p(ll n, ll p, ll pk) {
    prod[0] = 1;
    for (int i = 1; i <= pk; ++i)
        if (i % p) prod[i] = prod[i - 1] * i % pk;
        else prod[i] = prod[i - 1];
    ll rt = 1;
    for (; n; n /= p) {
        rt = rt * mpow(prod[pk], n / pk, pk) % pk;
        rt = rt * prod[n % pk] % pk;
    }
    return rt;
} // (n! without factor p) % p^k

```

6.17 PiCount* [cad6d4]

```

ll PrimeCount(ll n) { // n ~ 10^13 => < 2s
    if (n <= 1) return 0;
    int v = sqrt(n), s = (v + 1) / 2, pc = 0;
    vector<int> smalls(v + 1), skip(v + 1), roughs(s);
    vector<ll> larges(s);
    for (int i = 2; i <= v; ++i) smalls[i] = (i + 1) / 2;
    for (int i = 0; i < s; ++i) {
        roughs[i] = 2 * i + 1;
        larges[i] = (n / (2 * i + 1) + 1) / 2;
    }
    for (int p = 3; p <= v; ++p) {
        if (smalls[p] > smalls[p - 1]) {
            int q = p * p;
            ++pc;
            if (1LL * q * q > n) break;
            skip[p] = 1;
            for (int i = q; i <= v; i += 2 * p) skip[i] = 1;
            int ns = 0;
            for (int k = 0; k < s; ++k) {
                int i = roughs[k];
                if (skip[i]) continue;
                ll d = 1LL * i * p;
                larges[ns] = larges[k] - (d <= v ? larges[smalls[d] - pc] : smalls[n / d]) + pc;
                roughs[ns++] = i;
            }
            s = ns;
            for (int j = v / p; j >= p; --j) {
                int c =
                    smalls[j] - pc, e = min(j * p + p, v + 1);
                for (int i = j * p; i < e; ++i) smalls[i] -= c;
            }
        }
        for (int k = 1; k < s; ++k) {
            const ll m = n / roughs[k];
            ll t = larges[k] - (pc + k - 1);
            for (int l = 1; l < k; ++l) {
                int p = roughs[l];
                if (1LL * p * p > m) break;
                t -= smalls[m / p] - (pc + l - 1);
            }
            larges[0] -= t;
        }
        return larges[0];
    }
}

```

6.18 Discrete Log* [da27bf]

```

int DiscreteLog(int s, int x, int y, int m) {
    constexpr int kStep = 32000;
    unordered_map<int, int> p;
    int b = 1;
    for (int i = 0; i < kStep; ++i) {
        p[y] = i;
        y = 1LL * y * x % m;
        b = 1LL * b * x % m;
    }
    for (int i = 0; i < m + 10; i += kStep) {
        s = 1LL * s * b % m;
        if (p.find(s) != p.end()) return i + kStep - p[s];
    }
    return -1;
}
int DiscreteLog(int x, int y, int m) {
}

```

```

if (m == 1) return 0;
int s = 1;
for (int i = 0; i < 100; ++i) {
    if (s == y) return i;
    s = 1LL * s * x % m;
}
if (s == y) return 100;
int p = 100 + DiscreteLog(s, x, y, m);
if (fpow(x, p, m) != y) return -1;
return p;
}

```

6.19 Berlekamp Massey [3eb6fa]

```

template <typename T>
vector<T> BerlekampMassey(const vector<T> &output) {
    vector<T> d(SZ(output) + 1), me, he;
    for (int f = 0, i = 1; i <= SZ(output); ++i) {
        for (int j = 0; j < SZ(me); ++j)
            d[i] += output[i - j - 2] * me[j];
        if ((d[i] - output[i - 1]) == 0) continue;
        if (me.empty()) {
            me.resize(f + i);
            continue;
        }
        vector<T> o(i - f - 1);
        T k = -d[i] / d[f]; o.pb(-k);
        for (T x : he) o.pb(x * k);
        o.resize(max(SZ(o), SZ(me)));
        for (int j = 0; j < SZ(me); ++j) o[j] += me[j];
        if (i - f + SZ(he) >= SZ(me)) he = me, f = i;
        me = o;
    }
    return me;
}

```

6.20 Primes

```

/* 12721 13331 14341 75577 123457 222557
   556679 999983 109774749 1076767633 100102021
   999997771 1001010013 1000512343 987654361 999991231
   999888733 98789101 987777733 999991921 1010101333
   1010102101 10000000000039 1000000000000037
   2305843009213693951 4611686018427387847
   9223372036854775783 18446744073709551557 */

```

6.21 Theorem

Lucas's Theorem

For non-negative integer n, m and prime P ,
 $C(m, n) \bmod P = C(m/M, n/M) * C(m \% M, n \% M) \bmod P$
 $= \text{mult}_i(C(m_i, n_i))$
 where m_i is the i -th digit of m in base P .

Kirchhoff's theorem

$A_{ii} = \deg(i), A_{ij} = (i, j) \in E ? -1 : 0$
 Deleting any one row, one column, and call the $\det(A)$

Nth Catalan recursive function:

$C_0 = 1, C_{n+1} = C_n * 2(2n+1)/(n+2)$

Mobius Formula

$u(n) = 1$, if $n = 1$
 $(-1)^m$, 若 n 無平方數因數, 且 $n = p_1 * p_2 * p_3 * \dots * p_k$
 0 , 若 n 有大於 1 的平方數因數

-Property

1. (積性函數) $u(a)u(b) = u(ab)$
2. $\sum_{d|n} u(d) = [n == 1]$

Mobius Inversion Formula

if $f(n) = \sum_{d|n} g(d)$
 then $g(n) = \sum_{d|n} u(d) f(n/d)$
 $= \sum_{d|n} u(d) f(n/d)$

-Application

the number/power of $\gcd(i, j) = k$

-Trick

分塊, $O(\sqrt{n})$

Chinese Remainder Theorem (m_i 兩兩互質)

```

x = a_1 (mod m_1)
x = a_2 (mod m_2)
....
x = a_i (mod m_i)
construct a solution:
Let M = m_1 * m_2 * m_3 * ... * m_n
Let M_i = M / m_i

```

```

t_i = 1 / M_i
t_i * M_i = 1 (mod m_i)

```

solution $x = a_1 * t_1 * M_1 + a_2 * t_2 * M_2 + \dots + a_n * t_n * M_n + k * M$
 $= k * M + a_i * t_i * M_i$, k is positive integer.
 under mod M , there is one solution $x = a_i * t_i * M_i$

Burnside's lemma

$|G| * |X/G| = \sum (|X^g|)$ where $g \in G$

總方法數: 每一種旋轉下不動點的個數總和除以旋轉的方法數

6.22 Estimation

n	2	3	4	5	6	7	8	9	20	30	40	50	100
$p(n)$	2	3	5	7	11	15	22	30	62	75	60	44	2e5
n	100	1e3	1e6	1e9	1e12	1e15	1e18						
$d(i)$	12	32	240	1344	6720	26880	103680						
n	1	2	3	4	5	6	7	8	9	10	11	12	13
$\binom{2n}{n}$	2	6	20	70	252	924	3432	12870	48620	184756	7e5	2e6	1e7
n	2	3	4	5	6	7	8	9	10	11	12	13	
B_n	2	5	15	52	203	877	4140	21147	115975	7e5	4e6	3e7	

6.23 General Purpose Numbers

- Bernoulli numbers

$$B_0 = 1, B_1^\pm = \pm \frac{1}{2}, B_2 = \frac{1}{6}, B_3 = 0$$

$$\sum_{j=0}^m \binom{m+1}{j} B_j = 0, \text{EGF is } B(x) = \frac{x}{e^x - 1} = \sum_{n=0}^{\infty} B_n \frac{x^n}{n!}$$

$$S_m(n) = \sum_{k=1}^n k^m = \frac{1}{m+1} \sum_{k=0}^m \binom{m+1}{k} B_k^+ n^{m+1-k}$$

- Stirling numbers of the second kind Partitions of n distinct elements into exactly k groups.

$$S(n, k) = S(n-1, k-1) + k S(n-1, k), S(n, 1) = S(n, n) = 1$$

$$S(n, k) = \frac{1}{k!} \sum_{i=0}^k (-1)^{k-i} \binom{k}{i} i^n$$

$$x^n = \sum_{i=0}^n S(n, i) (x)_i$$

- Pentagonal number theorem

$$\prod_{n=1}^{\infty} (1 - x^n) = 1 + \sum_{k=1}^{\infty} (-1)^k \left(x^{k(3k+1)/2} + x^{k(3k-1)/2} \right)$$

- Catalan numbers

$$C_n^{(k)} = \frac{1}{(k-1)n+1} \binom{kn}{n}$$

$$C^{(k)}(x) = 1 + x [C^{(k)}(x)]^k$$

- Eulerian numbers

Number of permutations $\pi \in S_n$ in which exactly k elements are greater than the previous element. k, j : s.s.t. $\pi(j) > \pi(j+1), k+1, j$: s.s.t. $\pi(j) \geq j, k, j$: s.s.t. $\pi(j) > j$.

$$E(n, k) = (n-k)E(n-1, k-1) + (k+1)E(n-1, k)$$

$$E(n, 0) = E(n, n-1) = 1$$

$$E(n, k) = \sum_{j=0}^k (-1)^j \binom{n+1}{j} (k+1-j)^n$$

6.24 Tips for Generating Functions

- Ordinary Generating Function $A(x) = \sum_{i \geq 0} a_i x^i$

- $A(rx) \Rightarrow r^n a_n$
- $A(x) + B(x) \Rightarrow a_n + b_n$
- $A(x)B(x) \Rightarrow \sum_{i=0}^n a_i b_{n-i}$
- $A(x)^k \Rightarrow \sum_{i_1+i_2+\dots+i_k=n} a_{i_1} a_{i_2} \dots a_{i_k}$
- $x A(x)' \Rightarrow n a_n$
- $\frac{A(x)}{1-x} \Rightarrow \sum_{i=0}^n a_i$

- Exponential Generating Function $A(x) = \sum_{i \geq 0} \frac{a_i}{i!} x^i$

- $A(x) + B(x) \Rightarrow a_n + b_n$
- $A^{(k)}(x) \Rightarrow a_{n+k}$
- $A(x)B(x) \Rightarrow \sum_{i=0}^n \binom{n}{i} a_i b_{n-i}$
- $A(x)^k \Rightarrow \sum_{i_1+i_2+\dots+i_k=n} \binom{n}{i_1, i_2, \dots, i_k} a_{i_1} a_{i_2} \dots a_{i_k}$
- $x A(x) \Rightarrow n a_n$

- Special Generating Function

- $(1+x)^n = \sum_{i \geq 0} \binom{n}{i} x^i$
- $\frac{1}{(1-x)^n} = \sum_{i \geq 0} \binom{i}{n-1} x^i$

7 Polynomial

7.1 Fast Fourier Transform [5e2ea2]

```

const int maxn = 131072;
using cplx = complex<double>;
const cplx I = cplx(0, 1);
const double pi = acos(-1);
cplx omega[maxn + 1];

```

```

void prefft() {
    for (int i = 0; i <= maxn; ++i) omega[i] = exp(i * 2 * pi / maxn * I);
}

```

```

void bin(vector<cplx> &a, int n) {
    int lg;
    for (lg = 0; (1 << lg) < n; ++lg); --lg;
    vector<cplx> tmp(n);
    for (int i = 0; i < n; ++i) {
        int to = 0;
        for (int j = 0; (1 << j) <
              n; ++j) to |= (((i >> j) & 1) << (lg - j));
        tmp[to] = a[i];
    }
    for (int i = 0; i < n; ++i) a[i] = tmp[i];
}

void fft(vector<cplx> &a, int n) {
    bin(a, n);
    for (int step = 2; step <= n; step <= 1) {
        int to = step >> 1;
        for (int i = 0; i < n; i += step) {
            for (int k = 0; k < to; ++k) {
                cplx x = a[i
                    + to + k] * omega[maxn / step * k];
                a[i + to + k] = a[i + k] - x;
                a[i + k] += x;
            }
        }
    }
}

void ifft(vector<cplx> &a, int n) {
    fft(a, n);
    reverse(a.begin() + 1, a.end());
    for (int i = 0; i < n; ++i) a[i] /= n;
}

vector<int> multiply(const vector<
    int> &a, const vector<int> &b, bool trim = false) {
    int d = 1;
    while
        (d < max(a.size(), b.size())) d <= 1; d <= 1;
    vector<cplx> pa(d), pb(d);
    for (int i
        = 0; i < a.size(); ++i) pa[i] = cplx(a[i], 0);
    for (int i
        = 0; i < b.size(); ++i) pb[i] = cplx(b[i], 0);
    fft(pa, d); fft(pb, d);
    for (int i = 0; i < d; ++i) pa[i] *= pb[i];
    ifft(pa, d);
    vector<int> r(d);
    for (int
        i = 0; i < d; ++i) r[i] = round(pa[i].real());
    if (trim)
        while (r.size() && r.back() == 0) r.pop_back();
    return r;
}

```

Prime	Root	Prime	Root
7681	17	167772161	3
12289	11	104857601	3
40961	3	985661441	3
65537	3	998244353	3
786433	10	1107296257	10
5767169	3	2013265921	31
7340033	3	2810183681	11
23068673	3	2885681153	3
469762049	3	605028353	3

7.2 Number Theory Transform* [7d51db]

```

vector<int> omega;
void Init() {
    omega.resize(kN + 1);
    long long x = fpow(kRoot, (Mod - 1) / kN);
    omega[0] = 1;
    for (int i = 1; i <= kN; ++i) {
        omega[i] = 1LL * omega[i - 1] * x % kMod;
    }
}

void Transform(vector<int> &v, int n) {
    BitReverse(v, n);
    for (int s = 2; s <= n; s <= 1) {
        int z = s >> 1;
        for (int i = 0; i < n; i += s) {
            for (int k = 0; k < z; ++k) {
                int x = 1LL
                    * v[i + k + z] * omega[kN / s * k] % kMod;
                v[i + k + z] = (v[i + k] + kMod - x) % kMod;
                (v[i + k] += x) %= kMod;
            }
        }
    }
}

```

```

}
}
void InverseTransform(vector<int> &v, int n) {
    Transform(v, n);
    for (int i = 1; i < n / 2; ++i) swap(v[i], v[n - i]);
    const int kInv = fpow(n, kMod - 2);
    for (int i
        = 0; i < n; ++i) v[i] = 1LL * v[i] * inv % kMod;
}

```

7.3 Fast Walsh Transform* [c9cdb6]

```

/* x: a[j], y: a[j + (L >> 1)]
or: (y += x * op), and: (x += y * op)
xor: (x, y = (x + y) * op, (x - y) * op)
invop: or, and, xor = -1, -1, 1/2 */
void fwt(int *a, int n, int op) { //or
    for (int L = 2; L <= n; L <= 1)
        for (int i = 0; i < n; i += L)
            for (int j = i; j < i + (L >> 1); ++j)
                a[j + (L >> 1)] += a[j] * op;
}

const int N = 21;
int f[
    N][1 << N], g[N][1 << N], h[N][1 << N], ct[1 << N];
void
    subset_convolution(int *a, int *b, int *c, int L) {
    // c_k = \sum_{i | j = k, i & j = 0} a_i * b_j
    int n = 1 << L;
    for (int i = 1; i < n; ++i)
        ct[i] = ct[i & (i - 1)] + 1;
    for (int i = 0; i < n; ++i)
        f[ct[i]][i] = a[i], g[ct[i]][i] = b[i];
    for (int i = 0; i <= L; ++i)
        fwt(f[i], n, 1), fwt(g[i], n, 1);
    for (int i = 0; i <= L; ++i)
        for (int j = 0; j <= i; ++j)
            for (int x = 0; x < n; ++x)
                h[i][x] += f[j][x] * g[i - j][x];
    for (int i = 0; i <= L; ++i)
        fwt(h[i], n, -1);
    for (int i = 0; i < n; ++i)
        c[i] = h[ct[i]][i];
}

```

7.4 Polynomial Operation [869cb1]

```

#define
    fi(s, n) for (int i = (int)(s); i < (int)(n); ++i)
template<int MAXN, ll P, ll RT> // MAXN = 2^k
struct Poly : vector<ll> { // coefficients in [0, P)
    using vector<ll>::vector;
    static NTR<MAXN, P, RT> ntt;
    int n() const { return (int)size(); } // n() >= 1
    Poly(const Poly &p, int m) : vector<ll>(m) {
        copy_n(p.data(), min(p.n(), m), data());
    }
    Poly& irev() {
        return reverse(data(), data() + n()), *this;
    }
    Poly& isz(int m) { return resize(m), *this; }
    Poly& iadd(const Poly &rhs) { // n() == rhs.n()
        fi(0, n()) if
            (((*this)[i] += rhs[i]) >= P) (*this)[i] -= P;
        return *this;
    }
    Poly& imul(ll k) {
        fi(0, n()) (*this)[i] = (*this)[i] * k % P;
        return *this;
    }
    Poly Mul(const Poly &rhs) const {
        int m = 1;
        while (m < n() + rhs.n() - 1) m <= 1;
        Poly X(*this, m), Y(rhs, m);
        ntt(X.data(), m), ntt(Y.data(), m);
        fi(0, m) X[i] = X[i] * Y[i] % P;
        ntt(X.data(), m, true);
        return X.isz(n() + rhs.n() - 1);
    }
    Poly Inv() const { // (*this)[0] != 0, 1e5/95ms
        if (n() == 1) return {ntt.minv((*this)[0])};
        int m = 1;
        while (m < n() * 2) m <= 1;
        Poly Xi = Poly(*this, (n() + 1) / 2).Inv().isz(m);
        Poly Y(*this, m);
        ntt(Xi.data(), m), ntt(Y.data(), m);
        fi(0, m) {

```

```

    Xi[i] *= (2 - Xi[i] * Y[i]) % P;
    if ((Xi[i] % P) < 0) Xi[i] += P;
}
ntt(Xi.data(), m, true);
return Xi.isz(n());
}
Poly Sqrt()
const { // Jacobi((*this)[0], P) = 1, 1e5/235ms
if (n()
== 1) return {QuadraticResidue((*this)[0], P)};
Poly
X = Poly(*this, (n() + 1) / 2).Sqrt().isz(n());
return
X.iadd(Mul(X.Inv()).isz(n())).imul(P / 2 + 1);
}
pair<Poly, Poly> DivMod
(const Poly &rhs) const { // (rhs.)back() != 0
if (n() < rhs.n()) return {{0}, *this};
const int m = n() - rhs.n() + 1;
Poly X(rhs); X.irev().isz(m);
Poly Y(*this); Y.irev().isz(m);
Poly Q = Y.Mul(X.Inv()).isz(m).irev();
X = rhs.Mul(Q), Y = *this;
fi(0, n()) if ((Y[i] - X[i]) < 0) Y[i] += P;
return {Q, Y.isz(max(1, rhs.n() - 1))};
}
Poly Dx() const {
Poly ret(n() - 1);
fi(0,
ret.n()) ret[i] = (i + 1) * (*this)[i + 1] % P;
return ret.isz(max(1, ret.n()));
}
Poly Sx() const {
Poly ret(n() + 1);
fi(0, n())
ret[i + 1] = ntt.minv(i + 1) * (*this)[i] % P;
return ret;
}
Poly _tmul(int nn, const Poly &rhs) const {
Poly Y = Mul(rhs).isz(n() + nn - 1);
return Poly(Y.data() + n() - 1, Y.data() + Y.n());
}
vector<ll> _eval(const
vector<ll> &x, const vector<Poly> &up) const {
const int m = (int)x.size();
if (!m) return {};
vector<Poly> down(m * 2);
// down[1] = DivMod(up[1]).second;
// fi(2, m *
2) down[i] = down[i / 2].DivMod(up[i]).second;
down[1] = Poly(up[1])
.irev().isz(n()).Inv().irev()._tmul(m, *this);
fi(2, m * 2) down[i]
= up[i ^ 1]._tmul(up[i].n() - 1, down[i / 2]);
vector<ll> y(m);
fi(0, m) y[i] = down[m + i][0];
return y;
}
static vector<Poly> _tree1(const vector<ll> &x) {
const int m = (int)x.size();
vector<Poly> up(m * 2);
fi(0, m) up[m + i] = {(x[i] ? P - x[i] : 0), 1};
for (int i = m - 1; i
> 0; --i) up[i] = up[i * 2].Mul(up[i * 2 + 1]);
return up;
}
vector
<ll> Eval(const vector<ll> &x) const { // 1e5, 1s
auto up = _tree1(x); return _eval(x, up);
}
static Poly Interpolate(const vector
<ll> &x, const vector<ll> &y) { // 1e5, 1.4s
const int m = (int)x.size();
vector<Poly> up = _tree1(x), down(m * 2);
vector<ll> z = up[1].Dx()._eval(x, up);
fi(0, m) z[i] = y[i] * ntt.minv(z[i]) % P;
fi(0, m) down[m + i] = {z[i]};
for (int i = m -
1; i > 0; --i) down[i] = down[i * 2].Mul(up[i
* 2 + 1]).iadd(down[i * 2 + 1].Mul(up[i * 2]));
return down[1];
}
Poly Ln() const { // (*this)[0] == 1, 1e5/170ms
return Dx().Mul(Inv()).Sx().isz(n());
}
Poly Exp() const { // (*this)[0] == 0, 1e5/360ms

```

```

if (n() == 1) return {1};
Poly X = Poly(*this, (n() + 1) / 2).Exp().isz(n());
Poly Y = X.Ln(); Y[0] = P - 1;
fi(0, n())
if ((Y[i] = (*this)[i] - Y[i]) < 0) Y[i] += P;
return X.Mul(Y).isz(n());
}
// M := P(P - 1). If k >= M, k := k % M + M.
Poly Pow(ll k) const {
int nz = 0;
while (nz < n() && !(*this)[nz]) ++nz;
if (nz * min(k, (ll)n()) >= n()) return Poly(n());
if (!k) return Poly({1}, n());
Poly X(data() + nz, data() + nz + n() - nz * k);
const ll c = ntt.mpow(X[0], k % (P - 1));
return X.Ln().imul
(k % P).Exp().imul(c).irev().isz(n()).irev();
}
static ll
LinearRecursion(const vector<ll> &a, const vector
<ll> &coef, ll n) { // a_n = \sum c_j a_{n-j}
const int k = (int)a.size();
assert((int)coef.size() == k + 1);
Poly C(k + 1), W(Poly({1}, k), M = {0, 1});
fi(1, k + 1) C[k - i] = coef[i] ? P - coef[i] : 0;
C[k] = 1;
while (n) {
if (n % 2) W = W.Mul(M).DivMod(C).second;
n /= 2, M = M.Mul(M).DivMod(C).second;
}
ll ret = 0;
fi(0, k) ret = (ret + W[i] * a[i]) % P;
return ret;
}
};
#undef fi
using Poly_t = Poly<131072 * 2, 998244353, 3>;
template<> decltype(Poly_t::ntt) Poly_t::ntt = {};

```

7.5 Value Polynomial [96cde9]

```

struct Poly {
mint base; // f(x) = poly[x - base]
vector<mint> poly;
Poly(mint b = 0, mint x = 0): base(b), poly(1, x) {}
mint get_val(const mint &x) {
if (x >= base && x < base + SZ(poly))
return poly[x - base];
mint rt = 0;
vector<mint> lmul(SZ(poly), 1), rmul(SZ(poly), 1);
for (int i = 1; i < SZ(poly); ++i)
lmul[i] = lmul[i - 1] * (x - (base + i - 1));
for (int i = SZ(poly) - 2; i >= 0; --i)
rmul[i] = rmul[i + 1] * (x - (base + i + 1));
for (int i = 0; i < SZ(poly); ++i)
rt += poly[i] * ifac[i] * inegfac
[SZ(poly) - 1 - i] * lmul[i] * rmul[i];
return rt;
}
void raise() { // g(x) = sigma{base:x} f(x)
if (SZ(poly) == 1 && poly[0] == 0)
return;
mint nw = get_val(base + SZ(poly));
poly.pb(nw);
for (int i = 1; i < SZ(poly); ++i)
poly[i] += poly[i - 1];
}
};

```

7.6 Newton's Method

Given $F(x)$ where

$$F(x) = \sum_{i=0}^{\infty} \alpha_i (x - \beta)^i$$

for β being some constant. Polynomial P such that $F(P) = 0$ can be found iteratively. Denote by Q_k the polynomials such that $F(Q_k) = 0 \pmod{x^{2^k}}$, then

$$Q_{k+1} = Q_k - \frac{F(Q_k)}{F'(Q_k)} \pmod{x^{2^{k+1}}}$$

8 Geometry

8.1 Basic [e38806]

```

bool same
(double a, double b) { return abs(a - b) < eps; }

```



```

struct P {
    double x, y;
    P() : x(0), y(0) {}
    P(double x, double y) : x(x), y(y) {}
    P operator + (P b) { return P(x + b.x, y + b.y); }
    P operator - (P b) { return P(x - b.x, y - b.y); }
    P operator * (double b) { return P(x * b, y * b); }
    P operator / (double b) { return P(x / b, y / b); }
    double operator * (P b) { return x * b.x + y * b.y; }
    double operator ^ (P b) { return x * b.y - y * b.x; }
    double abs() { return hypot(x, y); }
    P unit() { return *this / abs(); }
    P rot(double o) {
        double c = cos(o), s = sin(o);
        return P(c * x - s * y, s * x + c * y);
    }
    double angle() { return atan2(y, x); }
};

struct L {
    // ax + by + c = 0
    double a, b, c, o;
    P pa, pb;
    L() : a(0), b(0), c(0), o(0), pa(), pb() {}
    L(P pa, P pb) : a(pa.y - pb.y), b(pb.x - pa.x),
        c(pa ^ pb), o(atan2(-a, b)), pa(pa), pb(pb) {}
    P project(P p) { return pa + (pb - pa).unit() * ((pb - pa) * (p - pa) / (pb - pa).abs()); }
    P reflect(P p) { return p + (project(p) - p) * 2; }
    double get_ratio(P p) { return (p - pa) * (pb - pa) / ((pb - pa).abs() * (pb - pa).abs()); }
    bool inside(P p) { return min(pa.x, pb.x) <= p.x && p.x <= max(pa.x, pb.x) && min(pa.y, pb.y) <= p.y && p.y <= max(pa.y, pb.y) && same(a * p.x + b * p.y, -c); }
};

bool SegmentIntersect(P p1, P p2, P p3, P p4) {
    if (max(p1.x, p2.x) < min(p3.x, p4.x) ||
        max(p3.x, p4.x) < min(p1.x, p2.x)) return false;
    if (max(p1.y, p2.y) < min(p3.y, p4.y) ||
        max(p3.y, p4.y) < min(p1.y, p2.y)) return false;
    return sign((p3 - p1) ^ (p4 - p1)) * sign((p3 - p2) ^ (p4 - p2)) <= 0 &&
        sign((p1 - p3) ^ (p2 - p3)) * sign((p1 - p4) ^ (p2 - p4)) <= 0;
}

bool parallel(L x, L y) { return same(x.a * y.b, x.b * y.a); }

P Intersect(L x, L y) { return P(-x.b * y.c + x.c * y.b, x.a * y.c - x.c * y.a) / (-x.a * y.b + x.b * y.a); }

```

8.2 KD Tree [375ca2]

```

namespace kdt {
int root, lc[maxn],
    rc[maxn], xl[maxn], xr[maxn], yl[maxn], yr[maxn];
point p[maxn];
int build(int l, int r, int dep = 0) {
    if (l == r) return -1;
    function<bool(const point &, const point &>> f = [dep](const point &a, const point &b) {
        if (dep & 1) return a.x < b.x;
        else return a.y < b.y;
    };
    int m = (l + r) >> 1;
    nth_element(p + l, p + m, p + r, f);
    xl[m] = xr[m] = p[m].x;
    yl[m] = yr[m] = p[m].y;
    lc[m] = build(l, m, dep + 1);
    if (~lc[m]) {
        xl[m] = min(xl[m], xl[lc[m]]);
        xr[m] = max(xr[m], xr[lc[m]]);
        yl[m] = min(yl[m], yl[lc[m]]);
        yr[m] = max(yr[m], yr[lc[m]]);
    }
    rc[m] = build(m + 1, r, dep + 1);
    if (~rc[m]) {
        xl[m] = min(xl[m], xl[rc[m]]);
        xr[m] = max(xr[m], xr[rc[m]]);
        yl[m] = min(yl[m], yl[rc[m]]);
        yr[m] = max(yr[m], yr[rc[m]]);
    }
}

```

```

return m;
}

bool bound(const point &q, int o, long long d) {
    double ds = sqrt(d + 1.0);
    if (q.x < xl[o] - ds || q.x > xr[o] + ds ||
        q.y < yl[o] - ds || q.y > yr[o] + ds) return false;
    return true;
}

long long dist(const point &a, const point &b) {
    return (a.x - b.x) * 1ll * (a.x - b.x) +
        (a.y - b.y) * 1ll * (a.y - b.y);
}

void dfs(
    const point &q, long long &d, int o, int dep = 0) {
    if (!bound(q, o, d)) return;
    long long cd = dist(p[o], q);
    if (cd != 0) d = min(d, cd);
    if ((dep & 1)
        && q.x < p[o].x || !(dep & 1) && q.y < p[o].y) {
        if (~lc[o]) dfs(q, d, lc[o], dep + 1);
        if (~rc[o]) dfs(q, d, rc[o], dep + 1);
    } else {
        if (~rc[o]) dfs(q, d, rc[o], dep + 1);
        if (~lc[o]) dfs(q, d, lc[o], dep + 1);
    }
}

void init(const vector<point> &v) {
    for (int i = 0; i < v.size(); ++i) p[i] = v[i];
    root = build(0, v.size());
}

long long nearest(const point &q) {
    long long res = 1e18;
    dfs(q, res, root);
    return res;
}
}

```

8.3 Sector Area [c41fb7]

```

// calc area of sector which include a, b
double SectorArea(P a, P b, double r) {
    double o = atan2(a.y, a.x) - atan2(b.y, b.x);
    while (o <= 0) o += 2 * pi;
    while (o >= 2 * pi) o -= 2 * pi;
    o = min(o, 2 * pi - o);
    return r * r * o / 2;
}

```

8.4 Half Plane Intersection [f7274e]

```

bool jizz(L l1, L l2, L l3) {
    P p = Intersect(l2, l3);
    return ((l1.pb - l1.pa) ^ (p - l1.pa)) < -eps;
}

bool cmp(const L &a, const L &b) {
    return same(
        a.o, b.o) ? (((b.pb - b.pa) ^ (a.pb - b.pa)) > eps) : a.o < b.o;
}

// available area for L l is (l.pb - l.pa) ^ (p - l.pa) > 0
vector<P> HPI(vector<L> &ls) {
    sort(ls.begin(), ls.end(), cmp);
    vector<L> pls(1, ls[0]);
    for (int i = 0; i < (int) ls.size(); ++i) if (!
        same(ls[i].o, pls.back().o)) pls.push_back(ls[i]);
    deque<int> dq; dq.push_back(0); dq.push_back(1);
#define meow(a, b, c
    ) while(dq.size() > 1u && jizz(pls[a], pls[b], pls[c]))
    for (int i = 2; i < (int) pls.size(); ++i) {
        meow(i, dq.back(), dq[dq.size() - 2]) dq.pop_back();
        meow(i, dq[0], dq[1]) dq.pop_front();
        dq.push_back(i);
    }
    meow(dq
        .front(), dq.back(), dq[dq.size() - 2]) dq.pop_back();
    meow(dq.back(), dq[0], dq[1]) dq.pop_front();
    if (dq.size() < 3u) return vector<P>(); // no solution or solution is not a convex
    vector<P> rt;
    for (int i = 0; i < (int) dq.size(); ++i) rt.push_back(
        Intersect(pls[dq[i]], pls[dq[(i + 1) % dq.size()]]));
    return rt;
}

```

8.5 Rotating Sweep Line [0411f0]

```

void rotatingSweepLine(vector<pair<int,int>> &ps){
    int n=int(ps.size());
    vector<int> id(n),pos(n);
    vector<pair<int,int>> line(n*(n-1)/2);
    int m=-1;
    for(int i=0;i<n;++i)for
        (int j=i+1;j<n;++j)line[++m]=make_pair(i,j); ++m;
    sort(line.begin(),line.end(),[&](const
        pair<int,int> &a,const pair<int,int> &b)->bool{
        if(ps
            [a.first].first==ps[a.second].first)return 0;
        if(
            [b.first].first==ps[b.second].first)return 1;
        return (double
            )(ps[a.first].second-ps[a.second].second)/(ps
            [a.first].first-ps[a.second].first) < (double
            )(ps[b.first].second-ps[b.second].second
            )/(ps[b.first].first-ps[b.second].first);
        });
    for(int i=0;i<n;++i)id[i]=i;
    sort(id.begin(),id.end(),[&](const
        int &a,const int &b){ return ps[a]<ps[b]; });
    for(int i=0;i<n;++i)pos[id[i]]=i;

    for(int i=0;i<n;++i){
        auto l=line[i];
        // meow
        tie(pos[l.first],pos[l.second],
            id[pos[l.first]],id[pos[l.second]])=make_tuple
            (pos[l.second],pos[l.first],l.second,l.first);
    }
}

```

8.6 Triangle Center [4e8ee9]

```

Point TriangleCircumCenter(Point a, Point b, Point c) {
    Point res;
    double a1 = atan2(b.y - a.y, b.x - a.x) + pi / 2;
    double a2 = atan2(c.y - b.y, c.x - b.x) + pi / 2;
    double ax = (a.x + b.x) / 2;
    double ay = (a.y + b.y) / 2;
    double bx = (c.x + b.x) / 2;
    double by = (c.y + b.y) / 2;
    double r1 = (sin(a2) * (ax - bx) + cos(a2) * (by
        - ay)) / (sin(a1) * cos(a2) - sin(a2) * cos(a1));
    return Point(ax + r1 * cos(a1), ay + r1 * sin(a1));
}

Point TriangleMassCenter(Point a, Point b, Point c) {
    return (a + b + c) / 3.0;
}

Point TriangleOrthoCenter(Point a, Point b, Point c) {
    return TriangleMassCenter(a, b
        , c) * 3.0 - TriangleCircumCenter(a, b, c) * 2.0;
}

Point TriangleInnerCenter(Point a, Point b, Point c) {
    Point res;
    double la = len(b - c);
    double lb = len(a - c);
    double lc = len(a - b);
    res.x = (
        la * a.x + lb * b.x + lc * c.x) / (la + lb + lc);
    res.y = (
        la * a.y + lb * b.y + lc * c.y) / (la + lb + lc);
    return res;
}

```

8.7 Polygon Center [ee6ff0]

```

Point BaryCenter(vector<Point> &p, int n) {
    Point res(0, 0);
    double s = 0.0, t;
    for (int i = 1; i < p.size() - 1; i++) {
        t = Cross(p[i] - p[0], p[i + 1] - p[0]) / 2;
        s += t;
        res.x += (p[0].x + p[i].x + p[i + 1].x) * t;
        res.y += (p[0].y + p[i].y + p[i + 1].y) * t;
    }
    res.x /= (3 * s);
    res.y /= (3 * s);
    return res;
}

```

8.8 Maximum Triangle [3a6d38]

```

double ConvexHullMaxTriangleArea
    (Point p[], int res[], int chnum) {
    double area = 0, tmp;
    res[chnum] = res[0];
    for (int i = 0, j = 1, k = 2; i < chnum; i++) {
        while (fabs(Cross(p[
            res[j]] - p[res[i]], p[res[(k + 1) % chnum]] -
            p[res[i]])) > fabs(Cross(p[res[j]] - p[res[i]],
            p[res[k]] - p[res[i]]))) k = (k + 1) % chnum;
        tmp = fabs(Cross(
            p[res[j]] - p[res[i]], p[res[k]] - p[res[i]]));
        if (tmp > area) area = tmp;
        while (fabs(Cross(p[
            res[(j + 1) % chnum]] - p[res[i]], p[res[k]] -
            p[res[i]])) > fabs(Cross(p[res[j]] - p[res[i]],
            p[res[k]] - p[res[i]]))) j = (j + 1) % chnum;
        tmp = fabs(Cross(
            p[res[j]] - p[res[i]], p[res[k]] - p[res[i]]));
        if (tmp > area) area = tmp;
    }
    return area / 2;
}

```

8.9 Point in Polygon [0a9a66]

```

int pip(vector<P> ps, P p) {
    int c = 0;
    for (int i = 0; i < ps.size(); ++i) {
        int a = i, b = (i + 1) % ps.size();
        L l(ps[a], ps[b]);
        P q = l.project(p);
        if ((p - q).abs() < eps && l.inside(q)) return 1;
        if (same(ps[
            a].y, ps[b].y) && same(ps[a].y, p.y)) continue;
        if (ps[a].y > ps[b].y) swap(a, b);
        if (ps[a].y <= p.y && p.y <
            ps[b].y && p.x <= ps[a].x + (ps[b].x - ps[a].x
            ) / (ps[b].y - ps[a].y) * (p.y - ps[a].y)) ++c;
    }
    return (c & 1) * 2;
}

```

8.10 Circle [466c44]

```

struct C {
    P c;
    double r;
    C(P c = P(0, 0), double r = 0) : c(c), r(r) {}
};

vector<P> Intersect(C a, C b) {
    if (a.r > b.r) swap(a, b);
    double d = (a.c - b.c).abs();
    vector<P> p;
    if (same(a.r + b.r,
        d)) p.push_back(a.c + (b.c - a.c).unit() * a.r);
    else if (a.r + b.r > d && d + a.r >= b.r) {
        double o = acos
            ((sq(a.r) + sq(d) - sq(b.r)) / (2 * a.r * d));
        P i = (b.c - a.c).unit();
        p.push_back(a.c + i.rot(o) * a.r);
        p.push_back(a.c + i.rot(-o) * a.r);
    }
    return p;
}

double IntersectArea(C a, C b) {
    if (a.r > b.r) swap(a, b);
    double d = (a.c - b.c).abs();
    if (d >= a.r + b.r - eps) return 0;
    if (d + a.r <= b.r + eps) return sq(a.r) * acos(-1);
    double p = acos
        ((sq(a.r) + sq(d) - sq(b.r)) / (2 * a.r * d));
    double q = acos
        ((sq(b.r) + sq(d) - sq(a.r)) / (2 * b.r * d));
    return p * sq(a.r) + q * sq(b.r) - a.r * d * sin(p);
}

```

```

// remove second
    level if to get points for line (defalut: segment)
vector<P> CircleCrossLine(P a, P b, P o, double r) {
    double x = b.x - a.x, y = b.y - a.y, A = sq(x) + sq(y)
        , B = 2 * x * (a.x - o.x) + 2 * y * (a.y - o.y);
    double C = sq(a.x - o.x
        ) + sq(a.y - o.y) - sq(r), d = B * B - 4 * A * C;
    vector<P> t;
    if (d >= -eps) {
        d = max(0., d);
        double i = (-B - sqrt(d)) / (2 * A);

```

```

    double j = (-B + sqrt(d)) / (2 * A);
    if (i - 1.0 <= eps && i >=
        -eps) t.emplace_back(a.x + i * x, a.y + i * y);
    if (j - 1.0 <= eps && j >=
        -eps) t.emplace_back(a.x + j * x, a.y + j * y);
}
return t;
}
// calc area
// intersect by circle with radius r and triangle OAB
double AreaOfCircleTriangle(P a, P b, double r) {
    bool ina = a.abs() < r, inb = b.abs() < r;
    auto p = CircleCrossLine(a, b, P(0, 0), r);
    if (ina) {
        if (inb) return abs(a ^ b) / 2;
        return SectorArea(b, p[0], r) + abs(a ^ p[0]) / 2;
    }
    if (inb) return
        SectorArea(p[0], a, r) + abs(p[0] ^ b) / 2;
    if (p.size() == 2) return SectorArea(a, p[0], r)
        + SectorArea(p[1], b, r) + abs(p[0] ^ p[1]) / 2;
    else return SectorArea(a, b, r);
}
// for any triangle
double AreaOfCircleTriangle(vector<P> ps, double r) {
    double ans = 0;
    for (int i = 0; i < 3; ++i) {
        int j = (i + 1) % 3;
        double o = atan2
            (ps[i].y, ps[i].x) - atan2(ps[j].y, ps[j].x);
        if (o >= pi) o = o - 2 * pi;
        if (o <= -pi) o = o + 2 * pi;
        ans += AreaOfCircleTriangle
            (ps[i], ps[j], r) * (o >= 0 ? 1 : -1);
    }
    return abs(ans);
}

```

8.11 Tangent of Circles and Points to Circle [19eb58]

```

vector<L> tangent(C a, C b) {
#define Pij \
    P i = (b.c - a.c).unit() * a.r, j = P(i.y, -i.x); \
    z.emplace_back(a.c + i, a.c + i + j);
#define deo(I,J) \
    double d = (a
        .c - b.c).abs(), e = a.r I b.r, o = acos(e / d); \
    P i =
        (b.c - a.c).unit(), j = i.rot(o), k = i.rot(-o); \
    z.emplace_back(a.c + j * a.r, b.c J j * b.r); \
    z.emplace_back(a.c + k * a.r, b.c J k * b.r);
    if (a.r < b.r) swap(a, b);
    vector<L> z;
    if ((a.c - b.c).abs() + b.r < a.r) return z;
    else if (same((a.c - b.c).abs() + b.r, a.r)) { Pij; }
    else {
        deo(+,+);
        if (same(d, a.r + b.r)) { Pij; }
        else if (d > a.r + b.r) { deo(+,-); }
    }
    return z;
}

vector<L> tangent(C c, P p) {
    vector<L> z;
    double d = (p - c.c).abs();
    if (same(d, c.r)) {
        P i = (p - c.c).rot(pi / 2);
        z.emplace_back(p, p + i);
    } else if (d > c.r) {
        double o = acos(c.r / d);
        P i = (p - c.c).unit
            (), j = i.rot(o) * c.r, k = i.rot(-o) * c.r;
        z.emplace_back(c.c + j, p);
        z.emplace_back(c.c + k, p);
    }
    return z;
}

```

8.12 Area of Union of Circles [490636]

```

vector<pair<double, double>> CoverSegment(C &a, C &b) {
    double d = (a.c - b.c).abs();
    vector<pair<double, double>> res;
    if (same(a.r + b.r, d)) ;
    else if (d <= abs(a.r - b.r) + eps) {

```

```

        if (a.r < b.r) res.emplace_back(0, 2 * pi);
    } else if (d < abs(a.r + b.r) - eps) {
        double o = acos((sq(a.r) + sq(d) - sq(b
            .r)) / (2 * a.r * d)), z = (b.c - a.c).angle();
        if (z < 0) z += 2 * pi;
        double l = z - o, r = z + o;
        if (l < 0) l += 2 * pi;
        if (r > 2 * pi) r -= 2 * pi;
        if (l > r) res.emplace_back
            (l, 2 * pi), res.emplace_back(0, r);
        else res.emplace_back(l, r);
    }
    return res;
}
double CircleUnionArea
    (vector<C> c) { // circle should be identical
    int n = c.size();
    double a = 0, w;
    for (int i = 0; w = 0, i < n; ++i) {
        vector<pair<double, double>> s = {{2 * pi, 9}}, z;
        for (int j = 0; j < n; ++j) if (i != j) {
            z = CoverSegment(c[i], c[j]);
            for (auto &e : z) s.push_back(e);
        }
        sort(s.begin(), s.end());
        auto F = [&](double t) { return c[i].r * (c[i].r *
            t + c[i].c.x * sin(t) - c[i].c.y * cos(t)); };
        for (auto &e : s) {
            if (e.first > w) a += F(e.first) - F(w);
            w = max(w, e.second);
        }
    }
    return a * 0.5;
}

```

8.13 Minimum Distance of 2 Polygons [7eb8bb]

```

// p, q is convex
double TwoConvexHullMinDist
    (Point P[], Point Q[], int n, int m) {
    int YMinP = 0, YMaxQ = 0;
    double tmp, ans = 999999999;
    for (i =
        0; i < n; ++i) if (P[i].y < P[YMinP].y) YMinP = i;
    for (i =
        0; i < m; ++i) if (Q[i].y > Q[YMaxQ].y) YMaxQ = i;
    P[n] = P[0], Q[m] = Q[0];
    for (int i = 0; i < n; ++i) {
        while (tmp = Cross(
            Q[YMaxQ + 1] - P[YMinP + 1], P[YMinP] - P[YMinP
                + 1]) > Cross(Q[YMaxQ] - P[YMinP + 1], P[YMinP
                    ] - P[YMinP + 1])) YMaxQ = (YMaxQ + 1) % m;
        if (tmp < 0) ans = min(ans, PointToSegDist
            (P[YMinP], P[YMinP + 1], Q[YMaxQ]));
        else ans = min(ans, TwoSegMinDist(P[
            YMinP], P[YMinP + 1], Q[YMaxQ], Q[YMaxQ + 1]));
        YMinP = (YMinP + 1) % n;
    }
    return ans;
}

```

8.14 2D Convex Hull [65eaab]

```

bool operator<(const P &a, const P &b) {
    return same(a.x, b.x) ? a.y < b.y : a.x < b.x;
}
bool operator>(const P &a, const P &b) {
    return same(a.x, b.x) ? a.y > b.y : a.x > b.x;
}
#define crx(a, b, c) ((b - a) ^ (c - a))

vector<P> convex(vector<P> ps) {
    vector<P> p;
    sort(ps.begin(), ps.end(), [&](P a, P b) { return
        same(a.x, b.x) ? a.y < b.y : a.x < b.x; });
    for (int i = 0; i < ps.size(); ++i) {
        while (p.size() >= 2 && crx(p[p.size() -
            2], ps[i], p[p.size() - 1]) >= 0) p.pop_back();
        p.push_back(ps[i]);
    }
    int t = p.size();
    for (int i = (int)p.size() - 2; i >= 0; --i) {
        while (p.size() > t && crx(p[p.size() -
            2], ps[i], p[p.size() - 1]) >= 0) p.pop_back();
        p.push_back(ps[i]);
    }
}

```

```

    p.pop_back();
    return p;
}

int sgn(double
    x) { return same(x, 0) ? 0 : x > 0 ? 1 : -1; }

P isLL(P p1, P p2, P q1, P q2) {
    double a = crx(q1, q2, p1), b = -crx(q1, q2, p2);
    return (p1 * b + p2 * a) / (a + b);
}

struct CH {
    int n;
    vector<P> p, u, d;
    CH() {}
    CH(vector<P> ps) : p(ps) {
        n = ps.size();
        rotate(p.begin(), min_element(p.begin(), p.end()), p.end());
        auto t = max_element(p.begin(), p.end());
        d = vector<P>(p.begin(), next(t));
        u = vector<P>(t, p.end()); u.push_back(p[0]);
    }
    int find(vector<P> &v, P d) {
        int l = 0, r = v.size();
        while (l + 5 < r) {
            int L = (l * 2 + r) / 3, R = (l + r * 2) / 3;
            if (v[L] * d > v[R] * d) r = R;
            else l = L;
        }
        int x = l;
        for (int i = l + 1; i < r; ++i) if (v[i] * d > v[x] * d) x = i;
        return x;
    }
    int findFarest(P v) {
        if (v.y > 0 || v.y == 0 && v.x > 0) return
            ((int)d.size() - 1 + find(u, v)) % p.size();
        return find(d, v);
    }
}

P get(int l, int r, P a, P b) {
    int s = sgn(crx(a, b, p[l % n]));
    while (l + 1 < r) {
        int m = (l + r) >> 1;
        if (sgn(crx(a, b, p[m % n])) == s) l = m;
        else r = m;
    }
    return isLL(a, b, p[l % n], p[(l + 1) % n]);
}

vector<P> getLineIntersect(P a, P b) {
    int X = findFarest((b - a).rot(pi / 2));
    int Y = findFarest((a - b).rot(pi / 2));
    if (X > Y) swap(X, Y);
    if (sgn(crx(a, b, p[X])) * sgn(crx(a, b, p[Y])) < 0)
        return {get(X, Y, a, b), get(Y, X + n, a, b)};
    return {}; // tangent case falls here
}

void update_tangent(P q, int i, int &a, int &b) {
    if (sgn(crx(q, p[a], p[i])) > 0) a = i;
    if (sgn(crx(q, p[b], p[i])) < 0) b = i;
}

void bs(int l, int r, P q, int &a, int &b) {
    if (l == r) return;
    update_tangent(q, l % n, a, b);
    int s = sgn(crx(q, p[l % n], p[(l + 1) % n]));
    while (l + 1 < r) {
        int m = (l + r) >> 1;
        if (sgn(crx(q, p[m % n], p[(m + 1) % n])) == s) l = m;
        else r = m;
    }
    update_tangent(q, r % n, a, b);
}

int x = l;
for (int i = l + 1; i < r; ++i) if (v[i] * d > v[x] * d) x = i;
return x;
}

int findFarest(P v) {
    if (v.y > 0 || v.y == 0 && v.x > 0) return
        ((int)d.size() - 1 + find(u, v)) % p.size();
    return find(d, v);
}

P get(int l, int r, P a, P b) {

```

```

    int s = sgn(crx(a, b, p[l % n]));
    while (l + 1 < r) {
        int m = (l + r) >> 1;
        if (sgn(crx(a, b, p[m % n])) == s) l = m;
        else r = m;
    }
    return isLL(a, b, p[l % n], p[(l + 1) % n]);
}

vector<P> getIS(P a, P b) {
    int X = findFarest((b - a).spin(pi / 2));
    int Y = findFarest((a - b).spin(pi / 2));
    if (X > Y) swap(X, Y);
    if (sgn(crx(a, b, p[X])) * sgn(crx(a, b, p[Y])) <
        0) return {get(X, Y, a, b), get(Y, X + n, a, b)};
    return {};
}

void update_tangent(P q, int i, int &a, int &b) {
    if (sgn(crx(q, p[a], p[i])) > 0) a = i;
    if (sgn(crx(q, p[b], p[i])) < 0) b = i;
}

void bs(int l, int r, P q, int &a, int &b) {
    if (l == r) return;
    update_tangent(q, l % n, a, b);
    int s = sgn(crx(q, p[l % n], p[(l + 1) % n]));
    while (l + 1 < r) {
        int m = (l + r) >> 1;
        if (sgn(crx(q, p[m % n], p[(m + 1) % n])) == s) l = m;
        else r = m;
    }
    update_tangent(q, r % n, a, b);
}

bool contain(P p) {
    if (p.x < d[0].x || p.x > d.back().x) return 0;
    auto it
        = lower_bound(d.begin(), d.end(), P(p.x, -1e12));
    if (it->x == p.x) {
        if (it->y > p.y) return 0;
    } else if (crx(*prev(it), *it, p) < -eps) return 0;
    it = lower_bound(u.begin(), u.end(), P(p.x, 1e12), greater<P>());
    if (it->x == p.x) {
        if (it->y < p.y) return 0;
    } else if (crx(*prev(it), *it, p) < -eps) return 0;
    return 1;
}

bool get_tangent(P p, int &a, int &b) { // b -> a
    if (contain(p)) return 0;
    a = b = 0;
    int i
        = lower_bound(d.begin(), d.end(), p) - d.begin();
    bs(0, i, p, a, b);
    bs(i, d.size(), p, a, b);
    i = lower_bound(u.begin(), u.end(), p, greater<P>()) - u.begin();
    bs((int)d.size() - 1, (int)d.size() - 1 + i, p, a, b);
    bs((int)d.size() - 1 + i, (int)d.size() - 1 + u.size(), p, a, b);
    return 1;
}
};

```

8.15 3D Convex Hull [29e4a9]

```

double
    absvol(const P a, const P b, const P c, const P d) {
    return abs(((b-a)^(c-a))*(d-a))/6;
}

struct convex3D {
    static const int maxn=1010;
    struct T {
        int a,b,c;
        bool res;
        T() {}
        T(int a, int b, int c, bool res=1):a(a),b(b),c(c),res(res) {}
    };
    int n,m;
    P p[maxn];
    T f[maxn*8];
    int id[maxn][maxn];
    bool on(T &t, P &q) {
        return ((p[t.c]-p[t.b])^(p[t.a]-p[t.b]))*(q-p[t.a])>eps;
    }
    void meow(int q, int a, int b) {

```



```

1) LCA
2) dfs with in[u] = dft++, out[u] = dft++
3) ord[in[u]] = ord[out[u]] = u
4) bitset<MAXN> inset
*/
struct Query {
    int L, R, LBid, lca;
    Query(int u, int v) {
        int c = LCA(u, v);
        if (c == u || c == v)
            q.lca = -1, q.L = out[c ^ u ^ v], q.R = out[c];
        else if (out[u] < in[v])
            q.lca = c, q.L = out[u], q.R = in[v];
        else
            q.lca = c, q.L = out[v], q.R = in[u];
        q.Lid = q.L / blk;
    }
    bool operator<(const Query &q) const {
        if (LBid != q.LBid) return LBid < q.LBid;
        return R < q.R;
    }
};

void flip(int x) {
    if (inset[x]) sub(arr[x]); // TODO
    else add(arr[x]); // TODO
    inset[x] = ~inset[x];
}

void solve(vector<Query> query) {
    sort(ALL(query));
    int L = 0, R = 0;
    for (auto q : query) {
        while (R < q.R) flip(ord[++R]);
        while (L > q.L) flip(ord[--L]);
        while (R > q.R) flip(ord[R-]);
        while (L < q.L) flip(ord[L++]);
        if (~q.lca) add(arr[q.lca]);
        // answer query
        if (~q.lca) sub(arr[q.lca]);
    }
}

```

9.4 Additional Mo's Algorithm Trick

- Mo's Algorithm With Addition Only
 - Sort query same as the normal Mo's algorithm.
 - Foreach query $[l, r]$:
 - If $l/blk = r/blk$, brute-force.
 - If $l/blk \neq curL/blk$, initialize $curL := (l/blk + 1) \cdot blk$, $curR := curL - 1$
 - If $r > curR$, increase $curR$
 - decrease $curL$ to fit l , and then undo after answering
- Mo's Algorithm With Offline Second Time
 - Require: Changing answer $\equiv$ adding $f([l, r], r+1)$.
 - Require: $f([l, r], r+1) = f([l, r], r+1) - f([l, l], r+1)$.
 - Part1: Answer all $f([l, r], r+1)$ first.
 - Part2: Store $curR \rightarrow R$ for $curL$ (reduce the space to $O(N)$), and then answer them by the second offline algorithm.
 - Note: You must do the above symmetrically for the left boundaries.

9.5 Hilbert Curve [1274a3]

```

ll hilbert(int n, int x, int y) {
    ll res = 0;
    for (int s = n / 2; s; s >>= 1) {
        int rx = (x & s) > 0;
        int ry = (y & s) > 0;
        res += s * 111 * s * ((3 * rx) ^ ry);
        if (ry == 0) {
            if (rx == 1) x = s - 1 - x, y = s - 1 - y;
            swap(x, y);
        }
    }
    return res;
} // n = 2^k

```

9.6 Dynamic Convex Trick* [673ffd]

```

// only works for integer coordinates!! maintain max
struct Line {
    mutable ll a, b, p;
    bool operator
    <(const Line &rhs) const { return a < rhs.a; }
    bool operator<(ll x) const { return p < x; }
};

struct DynamicHull : multiset<Line, less<>> {
    static const ll kInf = 1e18;
    ll Div(ll a,
        ll b) { return a / b - ((a ^ b) < 0 && a % b); }
    bool isect(iterator x, iterator y) {

```

```

        if (y == end()) { x->p = kInf; return 0; }
        if (x
            ->a == y->a) x->p = x->b > y->b ? kInf : -kInf;
        else x->p = Div(y->b - x->b, x->a - y->a);
        return x->p >= y->p;
    }
    void addline(ll a, ll b) {
        auto z = insert({a, b, 0}), y = z++, x = y;
        while (isect(y, z)) z = erase(z);
        if (x != begin
            () && isect(--x, y)) isect(x, y = erase(y));
        while ((y = x) != begin
            () && (--x)->p >= y->p) isect(x, erase(y));
    }
    ll query(ll x) {
        auto l = *lower_bound(x);
        return l.a * x + l.b;
    }
};

```

9.7 All LCS* [78a378]

```

void all_lcs(string s, string t) { // 0-base
    vector<int> h(SZ(t));
    iota(ALL(h), 0);
    for (int a = 0; a < SZ(s); ++a) {
        int v = -1;
        for (int c = 0; c < SZ(t); ++c)
            if (s[a] == t[c] || h[c] < v)
                swap(h[c], v);
        // LCS[s[0, a], t[b, c]] =
        // c - b + 1 - sum([h[i] >= b] | i <= c)
        // h[i] might become -1 !!
    }
}

```

9.8 Adaptive Simpson* [4074b3]

```

template<typename Func, typename d = double>
struct Simpson {
    using pdd = pair<d, d>;
    Func f;
    pdd mix(pdd l, pdd r, optional<d> fm = {}) {
        d h = (r.X - l.X) / 2, v = fm.value_or(f(l.X + h));
        return {v, h / 3 * (l.Y + 4 * v + r.Y)};
    }
    d eval(pdd l, pdd r, d fm, d eps) {
        pdd m((l.X + r.X) / 2, fm);
        d s = mix(l, r, fm).second;
        auto [flm, sl] = mix(l, m);
        auto [fmr, sr] = mix(m, r);
        d delta = sl + sr - s;
        if (abs(delta)
            ) <= 15 * eps) return sl + sr + delta / 15;
        return eval(l, m, flm, eps / 2) +
            eval(m, r, fmr, eps / 2);
    }
    d eval(d l, d r, d eps) {
        return eval
            ({l, f(l)}, {r, f(r)}, f((l + r) / 2), eps);
    }
    d eval2(d l, d r, d eps, int k = 997) {
        d h = (r - l) / k, s = 0;
        for (int i = 0; i < k; ++i, l += h)
            s += eval(l, l + h, eps / k);
        return s;
    }
};

template<typename Func>
Simpson<Func> make_simpson(Func f) { return {f}; }

```

9.9 Simulated Annealing [de78c6]

```

double factor = 100000;
const int base = 1e9; // remember to run ~ 10 times
for (int it = 1; it <= 100000; ++it) {
    // ans:
    answer, nw: current value, rnd(): mt19937 rnd()
    if (exp(-(nw - ans
        ) / factor) >= (double)(rnd() % base) / base)
        ans = nw;
    factor *= 0.99995;
}

```

9.10 Tree Hash* [34aae5]

```

ull seed;
ull shift(ull x) {

```

```

x ^= x << 13;
x ^= x >> 7;
x ^= x << 17;
return x;
}
ull dfs(int u, int f) {
    ull sum = seed;
    for (int i : G[u])
        if (i != f)
            sum += shift(dfs(i, u));
    return sum;
}

```

9.11 Binary Search On Fraction [765c5a]

```

struct Q {
    ll p, q;
    Q go(Q b, ll d) { return {p + b.p*d, q + b.q*d}; }
};
bool pred(Q);
// returns smallest p/q in [lo, hi] such that
// pred(p/q) is true, and 0 <= p,q <= N
Q frac_bs(ll N) {
    Q lo{0, 1}, hi{1, 0};
    if (pred(lo)) return lo;
    assert(pred(hi));
    bool dir = 1, L = 1, H = 1;
    for (; L || H; dir = !dir) {
        ll len = 0, step = 1;
        for (int t = 0; t < 2 && (t ? step/=2 : step*=2);)
            if (Q mid = hi.go(lo, len + step);
                mid.p > N || mid.q > N || dir ^ pred(mid))
                t++;
            else len += step;
        swap(lo, hi = hi.go(lo, len));
        (dir ? L : H) = !!len;
    }
    return dir ? hi : lo;
}

```

9.12 Min Plus Convolution* [09b5c3]

```

// a is convex a[i+1]-a[i] <= a[i+2]-a[i+1]
vector<int> min_plus_convolution
(vector<int> &a, vector<int> &b) {
    int n = SZ(a), m = SZ(b);
    vector<int> c(n + m - 1, INF);
    auto dc = [&](auto Y, int l, int r, int jl, int jr) {
        if (l > r) return;
        int mid = (l + r) / 2, from = -1, &best = c[mid];
        for (int j = jl; j <= jr; ++j)
            if (int i = mid - j; i >= 0 && i < n)
                if (best > a[i] + b[j])
                    best = a[i] + b[j], from = j;
        Y(Y, l, mid - 1, jl, from), Y(Y, mid + 1, r, from, jr);
    };
    return dc(dc, 0, n - 1 + m - 1, 0, m - 1), c;
}

```

9.13 Bitset LCS [330ab1]

```

cin >> n >> m;
for (int i = 1; x; i <= n; ++i)
    cin >> x, p[x].set(i);
for (int i = 1; x; i <= m; i++) {
    cin >> x, (g = f) |= p[x];
    f.shiftLeftByOne(), f.set(0);
    ((f = g - f) ^= g) &= g;
}
cout << f.count() << '\n';

```

9.14 N Queens Problem [d1fcc]

```

void solve
(vector<int> &ret, int n) { // no sol when n=2,3
    if (n % 6 == 2) {
        for (int i = 2; i <= n; i += 2) ret.pb(i);
        ret.pb(3); ret.pb(1);
        for (int i = 7; i <= n; i += 2) ret.pb(i);
        ret.pb(5);
    } else if (n % 6 == 3) {
        for (int i = 4; i <= n; i += 2) ret.pb(i);
        ret.pb(2);
        for (int i = 5; i <= n; i += 2) ret.pb(i);
        ret.pb(1); ret.pb(3);
    } else {
        for (int i = 2; i <= n; i += 2) ret.pb(i);
    }
}

```

```

    for (int i = 1; i <= n; i += 2) ret.pb(i);
}
}

```

9.15 Digit DP [35e1ac]

```

int dfs
(int pos, int pre, int lead, int limit, int cnt){
    if(pos == -1) return max(cnt, 1ll);
    if(!lead && !limit && dp[pos][pre][cnt])
        [pos][pre][cnt] != -1) return dp[pos][pre][cnt];
    int lb = 0, rb = (limit ? s[pos] - '0' : 1), ret = 1;
    for(int i=lb; i<=rb; i++){
        ret += dfs(pos - 1, i, lead &
                    (i == 0), limit & (i == rb), cnt + (i==1));
        ret %= mod;
    }
    if(!lead && !limit) dp[pos][pre][cnt] = ret;
    return ret;
}
int calc(int x){
    s.clear();
    if(x==0) return 0;
    while(x){s += (x%2 + '0'); x/=2;}
    memset(dp, -1, sizeof dp);
    return dfs(sz(s)-1, 0, 1, 1, 0);
}

```

10 Python

10.1 Misc

```

# decimal
from decimal import *
setcontext(Context(prec
                  =MAX_PREC, Emax=MAX_EMAX, rounding=ROUND_FLOOR))
print(Decimal(input()) * Decimal(input()))
from fractions import Fraction
Fraction
    ("3.14159").limit_denominator(10).numerator # 22
# N*M
map(int, input().split())
arr2d = [[
    list(map(int, input().split())) for i in range(N)]
# a^b/M
pow(a, b, M)
# random
from random import *
randrange(L, R, step) # [L,R) L+k*step
randint(L, R) # int from [L,R]
choice(list) # pick 1 item from list
choices(list, k) # pick k item
shuffle(list)
Uniform(L, R) # float from [L,R]
# print
print(f"num: {num}, pi: {pi:.2f}, str: {text}")
print(1, 2, 3, 4, sep=" | ", end=" <-- END\n")
# file IO
r = open("filename.in")
a = r.read() # read whole content into one string
w = open("filename.out", "w")
w.write("123\n")
# IO redirection
import sys
sys.stdin = open("filename.in")
sys.stdout = open("filename.out", "w")
print("123", file=sys.stderr)
input_data = sys.stdin.buffer.read().split()
n = int(input_data[0])
arr = list(map(int, input_data[1 : n + 1]))
sys.set_int_max_str_digits(5000000)

```